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Micromechanical testing and property upscaling of planetary rocks: A critical review

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ABSTRACT

Knowledge of the mechanical behavior of planetary rocks is indispensable for space explorations. The scarcity of pristine samples and the irregular shapes of planetary meteorites make it difficult to obtain representative samples for conventional macroscale rock mechanics experiments (macro-RMEs). This critical review discusses recent advances in microscale RMEs (micro-RMEs) techniques and the upscaling methods for extracting mechanical parameters. Methods of mineralogical and microstructural analyses, along with non-destructive mechanical techniques, have provided new opportunities for studying planetary rocks with unprecedented precision and capabilities. First, we summarize several mainstream methods for obtaining the mineralogy and microstructure of planetary rocks. Then, nondestructive micromechanical testing methods, nanoindentation and atomic force microscopy (AFM), are detailed reviewed, illustrating the principles, advantages, influencing factors, and available testing results from literature. Subsequently, several feasible upscaling methods that bridge the micro-measurements of meteorite pieces to the strength of the intact body are introduced. Finally, the potential applications of planetary rock mechanics research to guiding the design and execution of space missions are environed, ranging from sample return missions and planetary defense to extraterrestrial construction. These discussions are expected to broaden the understanding of the microscale mechanical properties of planetary rocks and their significant role in deep space exploration.
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1. Introduction

As human exploration of the solar system and interstellar space intensifies, deep space exploration has become one of the hot spots in the international scientific community. Since the Soviet Union launched Luna 1 in 1959, more than 280 deep-space missions have been conducted to explore objects beyond Earth's orbit. These missions aim to uncover the formation and evolution of the solar system, identify potential habitats for extraterrestrial life, and assess resources on other celestial bodies [1–5]. These explorations have significantly contributed to scientific research, resource exploration, and space colonization initiatives.

Understanding the mechanical properties of planetary rocks is crucial for advancing deep space exploration missions. These properties are essential for tasks such as planetary sampling, establishing space research stations, planetary defense, and future space mining endeavors [6]. They provide valuable insights into the composition and evolutionary history of the planets [7]. By comprehending the compressive strength, elasticity, and toughness of meteorites, scientists can model asteroids, including formation, evolution, and response to impact events [8]. Additionally, studying these properties sheds light on how impact events influence the structure and composition of celestial bodies, aiding in the understanding of impact crater formation mechanisms on the Earth, Mars and the Moon [9–11]. It also helps to assess potential threats to Earth and develop deflection strategies [12,13]. Furthermore, investigating the physical and mechanical properties of meteorites offers clues about the origins of life and aids space mining by locating underground resources [14–16]. For successful space missions, it is imperative to consider the mechanical properties of target planetary rocks. For instance, landers and rovers must evaluate the strength, stability, and terrain of the target planetary

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surface to ensure successful landing and operation [17,18]. These mechanical properties guide in-situ resource utilization (ISRU) technologies, influencing material selection, processing methods, and overall resource utilization strategy [19].

Acquiring planetary rock samples presents challenges due to harsh environments. Planetary surfaces are companied with rugged terrain, extreme temperatures, and high radiation levels posing formidable obstacles to spacecraft and sampling equipment [20]. Thick deposits and surface cover the surface which obscure valuable rocks or soil beneath, necessitating specialized techniques for successful sample collection [21]. For instance, NASA's Perseverance rover faced difficulties drilling into Martian rocks due to rapid fragmentation, underscoring the complexities involved in sample collection [22]. Preserving and returning samples to Earth presents additional challenges such as ensuring the originality of samples during recovery, transportation, storage, and preventing contamination from the Earth's environment [23]. Furthermore, the scarcity of pristine samples and irregular shapes of planetary rocks complicate laboratory analysis [24]. Fig. 1c show the precious rock and dust sample from asteroid Ryugu in Hayabusa2 mission, which range in size from millimeters to sub-millimeters [25,28]. Obviously, traditional strength testing methods often don't work with these small, precious samples. The conventional macroscale rock mechanical experiments (macro-RMEs) require intact, standard shapes, which are difficult to prepare from planetary rock samples. Furthermore, the complex anisotropic mechanical behavior of the planetary rocks due to different chemical compositions and heterogeneous microstructure leads to complex failure characteristics, while macro-RMEs fail to provide insights into the microscale characteristics and the finer microstructures involved with impact history or presence of potential biosignatures [12,26,27].

Recent advances in microscale rock mechanics experiments (micro-RMEs) have offered new opportunities for studying planetary rocks with unprecedented precision. Techniques such as nanoindentation and atomic force microscopy (AFM) allow for nondestructive characterization of mechanical properties, providing vital data for future exploration. These methods require very small samples and are tolerant of the shapes, yielding strength, hardness, and deformation characteristics of the planetary rocks. Microscale techniques help understand the microstructural, compositional, and evolutional characteristics of rocks, essential for predicting macroscopic mechanical responses [29,30]. Correlative microscopy imaging and mechanical testing are often required to understand the relationships between a material's internal structure, chemical composition, and mechanical behavior [31].

This paper reviews the latest advancements in micro-RMEs for planetary rocks, detailing the technical principles, historical perspectives, and recent discoveries. It covers microscale mineralogical and structural analysis techniques, micromechanical testing of planetary rocks, and the coupling of microscale and macroscale rock mechanics parameters. Finally, the paper outlines the application prospects of microscale rock mechanics in future deep space exploration.

2. Methods of mineralogical and microstructural analyses

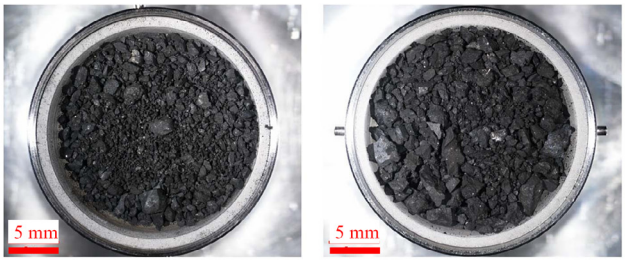
The microscopic structure and mineralogical compositions of extraterrestrial materials profoundly influence their mechanical behavior. Understanding the internal mineral constituents, crystalline structures, and morphological features of planetary rocks is essential. The microstructure of those rocks reveals the physical and chemical processes they experienced during formation, such as melting at high temperatures and pressures [32], cooling crystallization [33], and impact-induced venations [12]. This knowledge help in exploring their evolutionary history and formation processes. Various experimental techniques have been developed to analyze planetary rocks, providing valuable insights. This section summarizes the widely used methods for characterizing the microstructure and mineralogy of planetary rocks.

2.1. Microscopy

Microscopy, relying on visible light, polarized light, or electron as the means for interaction with the sample, is widely used to characterize the microstructure of planetary rocks [26,34,35]. Optical microscopes use visible light to illuminate specimens and observe surface features, veins, and mineral grains of extraterres-



(a) An image of Ryugu captured by optical navigation camera-telescopic



(b) Ryugu samples returned from Hayabusa2 (chamber A)



(c) Ryugu samples returned from Hayabusa2 (chamber C)

Fig. 1. Sample return missions targeted on the asteroid Ryugu (reprinted and adapted from [25,28]).

trial materials (Fig. 2a). Scanning electron microscopes (SEM) employ focused electron beams to scan specimen surfaces, creating images based on emitted secondary and backscattered electrons. It offers significantly higher magnifications and resolutions than optical microscopes, making it ideal for studying surface topography and morphology. SEM provides secondary electron (SE) images for surface topography and backscattered electron (BSE) images for compositional information (Fig. 2b). Transmission electron microscopes (TEM) use electron beams to produce high-resolution images of internal structures and compositions at the atomic level (Fig. 2c) [36]. It can be used to analyze the morphology of magnetite crystals associated with carbonate globules in meteorites, which have been proposed as potential biosignatures to discern signs of potential life [37].

Previous studies have used microscopes to examine surface features of planetary rocks such as grains [38] and mineral textures [39], or microscale impact features [40,41] like shock-induced fractures or melt pockets. For instance, Bischoff et al. [42] used optical microscope, and SEM to study the Jesenice meteorite observing its recrystallized texture and chondrule relics, while SEM revealed shock veins indicative of the shock history. Leroux et al. [40] used TEM to analyze the microstructure of pyroxene in the NWA 856 Martian meteorite and reconstructed its cooling and impact history.

2.2. Energy-dispersive spectroscopy (EDS)

Energy-dispersive spectroscopy (EDS), also known as energy-dispersive X-ray spectroscopy, is an analytical technique used to identify and quantify the elemental composition of materials. When high-energy electrons bombard a sample, the resulting X-rays emitted during the jump of electrons across different atomic orbitals are unique to each element. EDS detects these X-rays and measures their energy spectrum, correlating directly with the type and concentration of elements in the sample.

EDS works with various electron microscopy platforms, allowing for the analysis of small areas, such as individual grain or specific regions. It helps identify and classify minerals based on their morphology and elemental compositions, and provides elemental mapping of finite areas. For instance, EDS can distinguish between types of plagioclase feldspar by analyzing calcium-to-sodium ratios. SEM-EDS has been instrumental in studying shock metamorphism in rocks and minerals impacted by meteorites (Fig. 3) [31,43]. Through SEM-EDS analysis, researchers confirmed that shock fractures in quartz grains contain primarily amorphous silica, which is a classic indicator of shock metamorphism [44]. This distinction helps to separate fractures resulting from shock events from tectonic deformation lamellae. EDS is also utilized to investigate corrosion and weathering processes in rocks and minerals. By measuring the enrichment or depletion of elements during weathering, SEM-EDS assesses the degree of meteorite weathering. For instance, the Omani chondrites show significant Sr and Ba enrichment due to weathering and noticeable depletion of sulfur compared to unweathered chondrites [45]. Through analysis of surface morphology and changes in chemical composition, researchers can determine the mechanisms and rates of these processes, crucial for preserving the geological records (Fig. 4).

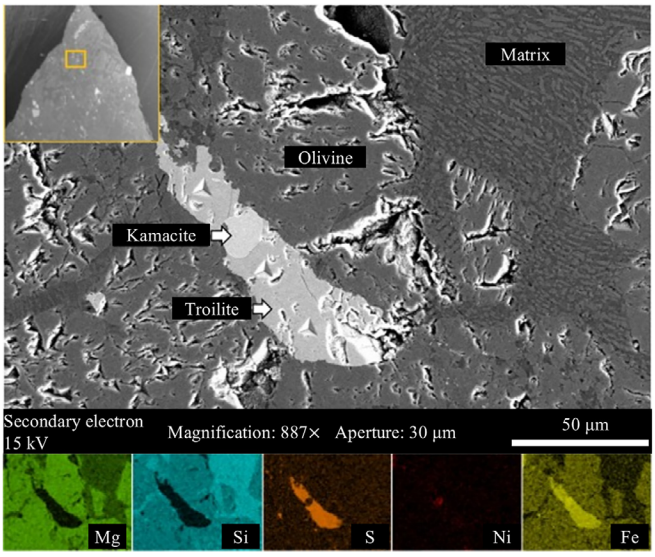
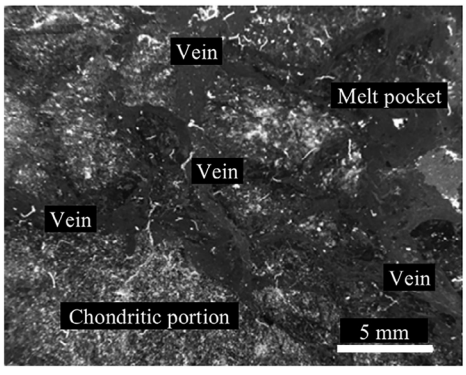
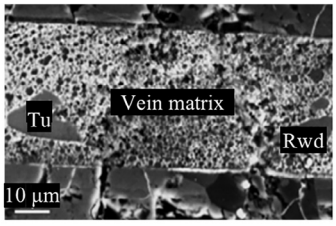


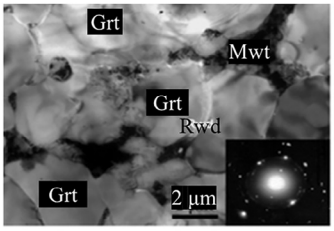
Fig. 3. SEM-EDS results show surface mapping and elemental distribution on a region of Aba Panu meteorites (adapted from [31]).



(a) Photograph of the dark melt veins in the chondrite

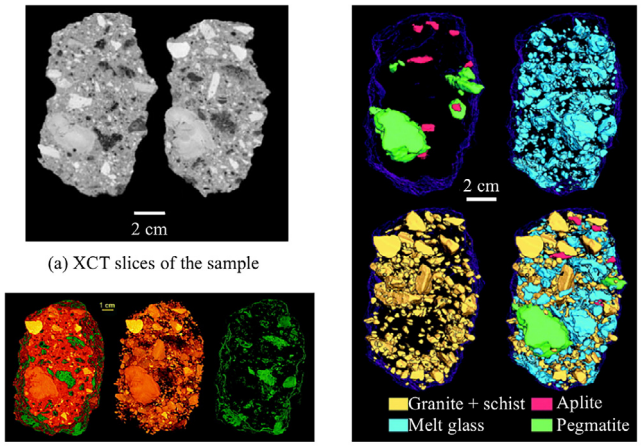


(b) BSE image of the melt vein



(c) TEM image of vein matrix

Fig. 2. Microscopy images of the chondrite.
Note: 'Tu' is tuite, "Rwd" is ringwoodite, "Grt" is garnet, and "Mwt" is magnesiowüstite (adapted from [36]).



(a) XCT slices of the sample

(b) 3D reconstruction of the clast population

(c) different types of rock clasts identified based on density differences

Fig. 4. High-resolution X-ray computed tomography examined the 3D distribution of clasts in suevites from the Bosumtwi craters, identifying different clast types based on density differences (adapted from [50]).

2.3. X-ray microscopy

X-ray microscopy techniques, such as X-ray computed tomography (XCT) and transmission X-ray microscopy (TXM), are non-destructive imaging methods that use X-rays to visualize the internal structure of a sample with high spatial resolution. XCT rotates the X-ray source and detectors around the sample to acquire two-dimensional (2D) images of multiple rotating angles, which are then reconstructed into a 3D image using computer algorithms. TXM images thin sections directly in transmission mode, achieving very high resolutions but requiring thin samples. This limits its applications in precious planetary rocks. XCT, which can image small planetary samples "as is", is more often used for planetary characterizations.

XCT is a potent tool for exploring the internal structure without compromising valuable samples. It has been used to investigate the morphology, internal structure, and properties of meteorites, interplanetary dust particles, lunar regolith, and materials related to 3D printing construction [46–50]. XCT reveals the 3D morphology of impact features and crystal growth, aiding in understanding how impact processes influence the rock's structure [31,51]. It is also useful for assessing physical characteristics such as porosity and density of rocks [52]. These XCT data can be utilized to construct numerical models for studying the micromechanical behavior of planetary rocks under various loads [48,53]. Combined with other techniques, XCT provides a comprehensive understanding of the physical and chemical properties of extraterrestrial materials. For example, X-ray combined with IR microtomography analyzed particles from the asteroid Itokawa and the Paris meteorite, providing 3D physical and chemical characterization, including the detection of mineral and organic phases and their spatial co-localization [54]. In-situ XCT scientific experimentation may be implemented in the near future. NASA is developing a prototype version of XCT (known as "Proto-XCT" or "iVOXEL") for non-destructive analysis in space. At present, the data analysis and processing are quite difficult due to the enormous data. A new generation of bright, highly coherent X-rays, combined with rapidly advancing machine learning, could bring a breakthrough to this status quo.

While XCT is generally non-destructive, high-energy X-rays can potentially alter the properties of samples, especially those with volatile constituents or sensitive to radiation. This can affect measurements of the rocks' original radiation history, including thermal and radiation history, orbits, terrestrial age, and cosmic ray exposure age. Sears et al. [55] noted that samples subjected to XCT receive radiation doses higher than natural exposure, altering the rocks' natural radiation history and original radiation record. Thus, caution is necessary when imaging rare or irreplaceable samples to prevent contamination or modification that could affect subsequent analyses.

2.4. X-ray diffraction (XRD)

XRD is an analytical technique used for determining the crystalline structure and chemical composition of non-amorphous minerals. It works on the principle that X-rays scattered by atoms in a crystal interfere constructively at specific angles, characteristic of the crystal structure. By measuring the angles and intensities of diffracted X-rays, a diffraction pattern is produced, which can be analyzed to identify mineral phases in the sample. XRD compares observed diffraction patterns with known patterns in a mineral to identify and estimate the abundance of different mineral. It can also detect amorphous phases or glasses, which are indicating the high-temperature processes like impact events or volcanic activity [56]. XRD can measure internal stress within crystal structures, providing insights into the geological history of the meteorites.

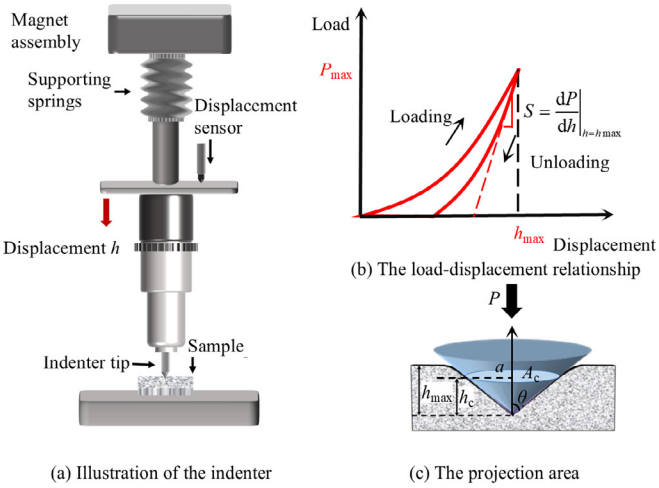
While XRD is the most effective method for analyzing crystalline minerals, it has limitations. Predominantly amorphous planetary samples or those with significant amounts of amorphous materials, such as glassy impact melts, can pose challenges in obtaining clear diffraction patterns. Some mineral phases may exist at low concentrations or have low X-ray scattering efficiency, making them difficult to be detected against the background noise [57]. Post-processing of diffraction data, including peak fitting, background subtraction, and phase quantification, requires appropriate methods and parameter selection, as improper data processing methods can yield incorrect results. Advances in computing and diffraction databases have made qualitative and quantitative analyses more straightforward and accessible through the Rietveld refinement method, now incorporated into many commercially available software packages [58].

2.5. Electron probe X-ray microanalysis (EPMA)

EPMA is a high-efficiency analytical instrument based on electron optics and X-ray spectroscopy, used for determining the chemical composition of materials with high spatial resolution. It utilizes a focused beam of electrons to excite atoms in a sample, causing them to emit characteristic X-rays. These X-rays, detected at specific wavelengths unique to each element, allow for the identification and quantification of elements. The X-rays can be counted based on the energy or wavelength of individual photons, leading to either energy-dispersive or wavelength-dispersive X-ray spectra (EDS or WDS). WDS typically offers higher resolution than EDS, because EDS can miscombine photons of similar energy into one type of X-ray. Unlike methods that analyze average chemical composition over a large range (such as wet chemical analysis or X-ray fluorescence), the micro-composition analysis of EPMA correlates micro-chemical composition with microstructure, enabling the study of areas as small as a few micrometers. This facilitates the examination of individual mineral grains or specific features within rocks. EPMA provides chemical data directly from the sample without extensive preparation, making it advantageous for delicate or rare samples. It can detect a wide range of elements, from beryllium (Be) to uranium (U), with particular sensitivity for elements with higher atomic numbers [59]. It can analyze solid-solution series in minerals, crucial for understanding crystallo-

chemical variations and geological processes. Proper sample surface preparation, requiring a highly polished surface, is crucial for quantitative analysis to avoid trapping or blocking electrons by rough surfaces.

EPMA ascertains the chemical composition of individual minerals within planetary rocks, offering insights into the rock's formation history and crystallization conditions. By scanning the electron beam across the sample, EPMA generates elemental maps revealing the distribution of different elements within the rock, highlighting features like mineral zoning or veinlets [48,60]. Combining EPMA data with corresponding BSE images enables the construction of microstructure-based representative volume elements (RVEs) for simulating the mechanical response of planetary rocks [61]. While EPMA is a powerful tool for chemical analysis of rocks and minerals, it does have limitations. Matrix effects can impact the accuracy of EPMA, as the presence of other elements in the sample influences the measured intensity of the element of interest [62]. Additionally, the high-energy electron beam can damage sensitive samples, particularly those containing volatile elements or those sensitive to radiation [63].



(a) Illustration of the indenter (b) The load-displacement relationship (c) The projection area

Fig. 5. Principles of nanoindentation tests.

3. Methods of nondestructive microscale mechanical experiments for planetary rocks

3.1. Nanoindentation

3.1.1. Principals of nanoindentation testing

Nanoindentation is a technique used to measure the mechanical properties of materials at the microscale and sub-microscale. By applying a controlled load to a sample with a smooth surface through a sharp and hard (e.g., diamond) tip of known geometry and monitoring the load-displacement relationship (as shown in Fig. 5a and b), mechanical parameters such as hardness and elastic modulus can be calculated using the Oliver-Pharr method [64]. The reduced modulus E_r is calculated by:

$$E_r = \frac{\sqrt{\pi}}{2\beta\sqrt{A_c}}S \tag{1}$$

$$S = \left.\frac{dP}{dh}\right|_{h=h_{max}} \tag{2}$$

where S is the stiffness calculated by the initial unloading slope; h_{max} the maximum penetration depth; and A_c the projection area (Fig. 5c). P is the load and P_{max} is the maximum load at the end of the loading stage.

The elastic modulus E and hardness H of the material can be determined as follows:

$$E = (1-\nu^2)\left[\frac{1}{E_r} - \frac{1-\nu_{tip}^2}{E_{tip}}\right]^{-1} \tag{3}$$

$$H = \frac{P_{max}}{A_c} \tag{4}$$

where E_{tip} and ν_{tip} are the elastic modulus and Poisson's ratio of the indenter tip; and ν the Poisson's ratio of the tested specimen.

Nanoindentation can be used to study the time-dependent creep behavior of materials. The creep characteristics are assessed by maintaining a constant load for a preset period of time and then analyzing the change in indentation depth over time. Methods such as the crack-length-based method, energy analysis method, micro-cantilever method, and micro-pillar splitting method can be used to evaluate the fracture toughness of materials by analyzing the crack length and energy consumption [65]. The elastic and plastic deformation behavior of materials can be obtained by the load-displacement curves [66]. Grid nanoindentation and statistical deconvolution are used in the test and data analysis [67].

3.1.2. Advantages of nanoindentation technique

Nanoindentation is a powerful technique used to study the mechanical properties of materials at the microscale. It offers several advantages over traditional macroscopic testing method, such as allowance of different rock shapes, minimal sample damage, and enabling multiple measurements without compromising the specimen's integrity [68]. These make nanoindentation particularly beneficial for precious or non-renewable sample like carbonaceous chondrite, lunar, and Martian regolith. After testing, the samples can be reused for other analyses, enhancing sample utilization efficiency. Its high spatial resolution allows researchers to probe localized features within individual grains. This is crucial for understanding the structure-property relationships in complex composite materials. Besides hardness and elastic modulus, nanoindentation provides other mechanical parameters such as yield strength, fracture toughness, creep behaviors, and energy dispersion [65]. The tests are straightforward and can generate substantial data quickly, making them suitable for rapid material assessments.

Combining nanoindentation with SEM, EDS, and XRD is a powerful approach to investigating the mechanical properties and microstructural characteristics at the nanoscale. For instance, Liu et al. [69] used this combined approach to identify different mineral groups within shale and determine their individual mechanical properties. SEM-EDS confirmed the mineralogy and provided spatially resolved elemental mapping, while XRD offered bulk mineralogical characterization. Tang et al. [70] utilized nanoindentation with the TESCAN integrated mineral analyzer (TIMA) to obtain the mechanical properties of meteorite minerals. TIMA is an automated quantitative analysis system for rocks and minerals based on SEM and EDS. Based on the digital mineralographic images obtained by TIMA, nanoindentation can measure the mechanical responses of individual phases in rocks.

3.1.3. Applications of nanoindentation in planetary rocks

Nanoindentation has been applied in previous investigations of lunar dust, meteorites, Martian rocks, and other geological samples to better understand their physical behavior and response. The technique can distinguish different mineral phases within a sample

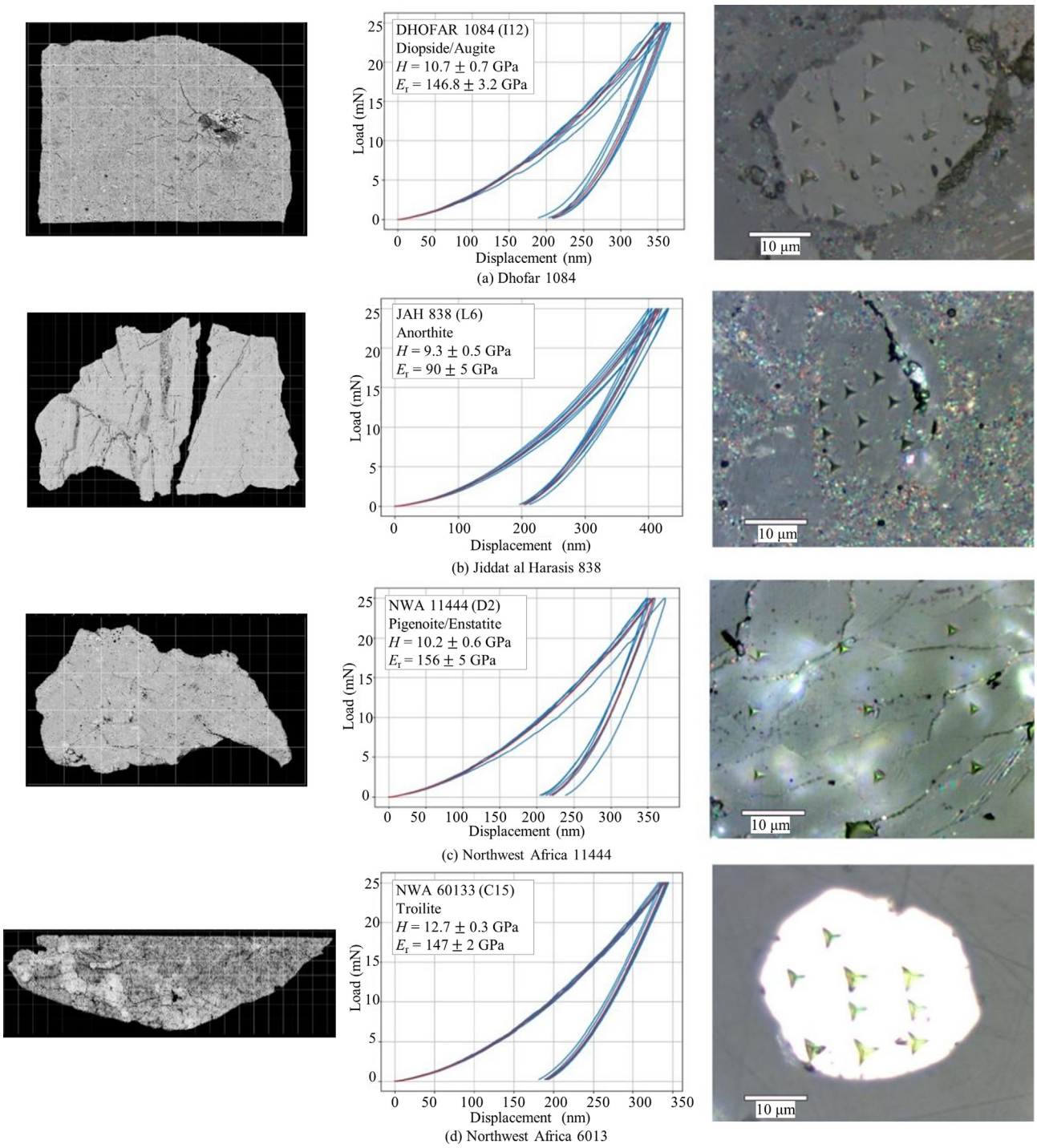


Fig. 6. Mechanical properties of lunar meteorites Dhofar 1084, JaH 838, NWA 11444, and HED meteorite NWA 6013.
Note: Images show the photos (left), load–displacement curves (mid) and residual indents (right). Images are adapted from [16].

based on their mechanical responses, as different phases exhibit distinct load-displacement curves (Fig. 6). Combined with microstructure and mineralogy analysis methods (such as SEM, EDS, or TIMA), nanoindentation can precisely measure the mechanical properties of specific mineral particles. This combination provides insights into the micromechanical behavior of minerals, crucial for understanding the macroscopic mechanical properties of rocks. For example, it can investigate how glassy agglutinations and metal particles in lunar dust influence the overall mechanical performance. Table 1 summarizes the available data from previous nanoindentation tests on planetary and terrestrial rocks, including asteroid meteorites, lunar regoliths, Martian meteorites, and lunar and Mars simulants.

On asteroid meteorites. Nanoindentation techniques have been successfully applied to study micron-scale particles collected from sample return missions and observed meteorites, providing valuable information on the physical properties of materials that form asteroids. Tested asteroid meteorites include Aba Panu [31], GRO 85209 [71,72], NWA 13618 [70], HaH 346 [73], asteroid 25,143 Itokawa [74], Chelyabinsk [12,75,76], Tarda [77], NWA

Table 1
Overview of mechanical properties of planetary rocks obtained by nanoindentation.

Name	Classification	Load (mN)	Mineral phases	Elastic modulus (GPa)	Hardness (GPa)
Asteroid meteorites					
Aba Panu [31]	L3 chondrite		Olivine	202.7±10.2	16.4±0.8
			Pyroxene	175.9±10.8	12.7±1.0
			Chromite	153.1±9.4	16.7±0.9
			Kamacite	211.3±9.4	3.3±0.1
			Troilite	108.0±3.4	3.9±0.2
GRO 85209 [71]	L3 chondrite		Pyroxene	95±3	
			Fe-Ni	178±6	
GRO 85209 [72]	L3 chondrite		Olivine	127±5	
			Pyroxene	123±5	
			Plagioclase	85±5	
			Chondrule	83±5	
			Fe-Ni	156±5	
NWA 13618 [70]	L4 chondrite	4	Olivine	116.73	
			Pyroxene	101.77	
			Fe-Ni	87.24	
			Feldspar	70.74	
HaH 346 [73]	L6 chondrite		Olivine	124.19±16.69	
			Pyroxene	106.63±13.78	
			Plagioclase	45.86±12.95	
			Troilite	93.36±6.07	
			Kamacite	150.96±7.54	
			Merrillite	40.28±11.22	
Itokawa (#14) [74]	LL4 chondrites	5	Mixture	83.01±0.12	8.01±0.01
		10		86.00±0.04	9.00±0.014
Itokawa (#23) [74]	LL4 chondrites	5	Mixture	111.01±0.22	10.01±0.02
		10		101.00±0.05	10.00±0.01
Itokawa (#47) [74]	LL4 chondrites	5	Mixture	86.01±0.13	13.01±0.03
		10		82.00±0.14	11.00±0.02
Chelyabinsk [75]	LL5 chondrites	500	Light-colored	65±11	9.2±2.4
			Dark-colored	65±15	11±4
			Impact melt	77±7	11.8±2.3
Chelyabinsk [12]	LL5 chondrites	20	Olivine	136±5	13.6±0.9
			Pyroxene	122±11	11.9±2.2
			Kamacite	127±16	3.58±0.24
			Taenite	82±6	3.05±0.29
			Troilite	71±8	5.1±0.8
			Chromite	131±3	15.9±1.3
Tarda [77]	Carbonaceous chondrites	1.8	Phase 1	18.12±1.36	0.88±0.11
			Phase 2	26.30±4.79	1.00±0.25
			Phase 3	39.45±7.45	2.42±0.56
			Phase 4	59.20±12.44	4.01±1.58
NWA 6013 [16]	Asteroidal achondrites, basaltic		Chromite	147	17.3
			Troilite	147	12.7
			Feldspar	88	10.6
			Pyroxene	134	12.6
NWA 7983[78]	Asteroidal achondrites,ureilites		micrometer-sizedlonsdaleite		40–65
			nanocrystalline lonsdaleite		About 40
Seymchan [33]	Stony-irons, Pallasite	7	Kamacite	241	3.55
			Taenite	65	3.98
			Plessite	219	4.22
		1	Tetrataenite	237	8.43
			Cloudy zone	208	7.23
Taza (NWA 859) [29]	Iron meteorites	15	Kamacite	116.73	2.555
			Plessite	185.8	2.879
			Tetrataenite	152.6	2.390
			Cloudy Zone	186.6	4.344
			Schreibersite	190.8	11.03
Martian rocks and simulants					
NWA 12564 [82]	Martian (shergottite)		Clinopyroxene	137.55±7.44	
			Plagioclase	74.13±0.93	
			Magnetite	103.90±3.32	
			Maskelynite	94.94±4.31	
			Chlorapatite	42.67±2.17	
MMS [84]	Mars Simulants		Sodium silicate composite	31.4±3.6	1.6±0.3
			Cement composite	21.3±2.4	0.80±0.19
MGS-1 [83]	Mars Simulants	500	Matrix	18.83	0.29
MGS-1S [83]	Mars Simulants	500	Matrix	25.24	0.34
			Aggregate	76.55	7.08
MMS-1S [83]	Mars Simulants	500	Matrix	21.74	0.49
			Aggregate	39.43	3.3
Lunar rocks, regolith, and simulants					
NWA 12008 [80]	Lunar (mare basalt)	100	Olivine	110±32	7.6±2.2
			Pyroxene	91±12	7.1±1.3

(*continued on next page*)

Table 1 (*continued*)

Name	Classification	Load (mN)	Mineral phases	Elastic modulus (GPa)	Hardness (GPa)
			Ilmenite	91±13	8.4±0.6
			Maskelinite	86±34	7.2±2.8
			Plagioclase	74	6.9
Dhofar 1084 [16]	Lunar (anorthositic breccia)	25	Feldspar	92–99	8.8–9.6
			Pyroxene	146–149	10.7–14.3
			Spinel	154	10.6
JaH 838 [16]	Lunar (mingled breccia)	25	Feldspar	79–90	7.4–10.6
			Olivine	113–139	8.8–10.2
			Calcite	74	2.8
			Pyroxene	134	11
			Spinel	116	9.6
NWA 11444 [16]	Lunar (melt breccia)	25	Feldspar	92	9
			Olivine	156–174	11.7–12.5
			Calcite	63	2.4
			Pyroxene	135–162	9.9–12.3
			Ilmenite	134	9.5
Chang'e-5 lunar sample [47]	Lunar regolith (BYX001GP)	10 mN, 5 s	Olivine	142.04	12.00
			Imenite	122.64	9.53
			Matrix	57.39	6.26
			Pyroxene	71.35	9.73
		10 mN, 20 s	Olivine	134.3	11.47
			Ilmenite	127.85	9.91
			Matrix	54.37	5.38
			Pyroxene	63.09	8.52
		50 mN, 5 s	Olivine	119.06	10.12
			Ilmenite	103.00	7.43
			Matrix	40.00	2.74
			Pyroxene	44.77	6.12
Chang'e-5 lunar sample [81]	Lunar glassy particles			49.1±3.5	7.4±0.5
				81.7±4.5	7.7±0.3
FJS-1 [90]	Lunar simulants	3	Anorthite	59.4–86.6	
			Augite	94–160	

6013 [16], NWA 7983 [78], Seymchan [33], and NWA 859 [29]. This mechanical data helps predict the response of asteroids to impact events and aids in the design of future asteroid mining or deflection strategies, such as the Asteroid Impact and Deflection Assessment (AIDA) mission [13,79]. For instance, the Chelyabinsk meteorite, a highly shocked, low porosity, ordinary chondrite, is similar to S- or Q-type asteroids, making it a valuable proxy for studying these types of celestial bodies [30,76]. Nanoindentation revealed a low fracture toughness in its shock-melt veins, indicating potential structural weaknesses that could promote material ejection after an impact, thus increasing momentum multiplication [12,75]. Tanbakouei et al.[74] used nanoindentation to study micro-sized particles from asteroid 25,143 Itokawa, providing insights into the cohesive and mechanical response of the surface regolith, which could help quantify the efficiency of crater excavation.

On lunar regolith, Martian rocks, and their simulants. The Apollo missions brought back a wealth of lunar samples, and nanoindentation has been an invaluable tool in the analysis of these samples [3]. By examining the mechanical properties of lunar rocks, researchers gain insights into the Moon's thermal history, interior nature, and surface processes over geological time. This information is essential for planning future lunar missions and potential lunar resource utilization. Research has focused on the mechanical properties of lunar meteorites [16,80], lunar glasses [81], and lunar regolith [47]. Grèbol-Tomàs et al. [80] revealed that lunar basalt meteorites NWA 12008 are softer than their terrestrial equivalents. Lunar glasses, originating mainly from meteorite impacts, have undergone extensive weathering and aging over millions to billions of years while maintaining their glassy nature. Understanding the aging of these materials is crucial for developing high-performance materials for radiation protection and advanced space explorations. Chen et al. [81] studied long-term aging effects on glassy particles from the Chang'e-5 lunar regolith using nanoindentation, finding significant increases in modulus (greater than 73.5%) and much weaker effects on hardness due to aging. They concluded that the aging effects of lunar glassy particles are primarily due to their complex compositions and intrinsic structures. Nie et al. [47] conducted a series of nondestructive microscopic tests on Chang'e-5 lunar samples, measuring properties like surface roughness and Young's modulus, to predict interparticle friction coefficients of lunar regolith particles.

So far, Martian meteorites are the only rock samples obtained from Mars. Compared to orbiting probes and rovers, these meteorites provide more detailed and complete information on the mineralogy of Martian rocks and allow for mechanical tests. Due to the destructiveness of traditional rock mechanics experiments and the relative scarcity of samples, there are few measurements of the mechanical properties of Martian meteorites. The only available literature on mechanical characterization based on nanoindentation comes from Zhang et al. [82]. They used nanoindentation tests to investigate the mechanical properties of minerals in the NWA12564 Martian meteorite, including clinopyroxene, plagioclase, magnetite, maskelynite, and chlorapatite, obtaining their Young's modulus and hardness.

Various simulants are used in experiments instead of real planetary rocks due to the accessibility limitations [32,83]. Real planetary rocks, such as lunar regolith, Martian rocks, and asteroid meteorites, are scarce and expensive to obtain because of the high cost of sample return missions. Thus, simulants produced on Earth are used for research and development purposes, which are microstructural and mineralogically similar to real planetary rocks. NASA recommends indigenous Martian materials such as crushed basaltic igneous rocks, sodium silicate, and magnesium oxide or basalt-based cements [84]. Mars simulants like JSC Mars-1, MMS, MGS-1, and their updated generations have been investigated in

experiments [83–85]. Several lunar regolith simulants, such as MLS-1, JSC-1, MKS-1, FJS-1, and so on, mimic the properties of actual lunar soil for research and testing [86,87].

There are several advantages to using these simulants. First, they are economically and readily available in large quantities, enabling extensive and comparative testing. For example, knowing whether the wheels on a rover have enough traction on a particular terrain requires extensive testbeds filled with simulants [88]. These soil analogs are also useful in calibrating scientific instruments on rovers [89]. Second, simulants can be formulated to have consistent compositions and properties, allowing for repeatable and controlled experiments, which validates results. Additionally, working with real planetary dusts poses challenges related to containment and environmental risks. Zhang et al. [90] investigated the application of spark plasma sintering (SPS) technology in consolidating lunar regolith simulants FJS-1, finding that the external pressure and heating rate are key factors influencing the elastic modulus of anorthite and augite. Zheng and Qiao [32] studied the mechanical properties of bulk glass made from melted lunar regolith simulant, obtaining nanohardness, Vickers hardness, elastic modulus, and fracture toughness of the material, which provide insights for in-situ fabrication of construction materials for future lunar outposts.

3.2. Atomic force microscopy (AFM)

3.2.1. Principles of AFM

Atomic force microscopy (AFM) is a high-resolution nanoscale characterization technique that provides detailed information about the surface topography and mechanical properties of materials. AFM operates using a highly sensitive microcantilever with a nanoscale tip mounted at its end. One end of the microcantilever is fixed, while the other interacts with the sample surface. As the tip interacts with the sample, the microcantilever bends or vibrates, and these changes are detected and recorded by a laser beam and photodetector. Analyzing this data reveals the sample's surface morphology, mechanical properties, and surface dynamics [91]. AFM has several operating modes, including contact mode, non-contact mode, and tapping mode [92]. Contact mode involves the AFM tip continuously touching the sample surface, allowing for fast mapping but potentially causing sample damage. Non-contact mode features the tip oscillating near the sample surface without direct contact, which is beneficial for studying soft or delicate samples such as biological membranes and polymers. Tapping mode combines the benefits of both, with the tip intermittently contacting the surface, providing high-resolution images with reduced sample alteration. This makes it suitable for a wide range of materials. In addition to these above basic modes, AFM has extra modes such as phase imaging mode and force modulation mode. These can provide additional information about the sample surface, such as adhesion, elasticity, and dissipation [93].

PeakForce quantitative nano-mechanical mapping (PeakForce QNM™) is an advanced mode of AFM that enhances traditional tapping mode by providing quantitative nanomechanical mapping of surfaces [94]. In PeakForce Tapping, the force exerted by the cantilever on the sample is precisely controlled and measured throughout scanning by maintaining a constant deflection, which corresponds to a constant force. This force exerted by the cantilever can be used to calculate the local elastic modulus (or other mechanical properties) of the sample by analyzing the force curves at each image pixel. The PeakForce QNM uses these force curves to separate the elastic and inelastic components, mapping the sample's mechanical properties. Fig. 7 illustrates the stages of recording the force response between the AFM probe and the sample surface. Stage 1 has no interaction force. In stage 2, van der Waals (vdW) attraction forces bend the cantilever as the tip approaches the surface. Stage 3 sees increased force upon contact until a preset maximum is reached. In stage 4, the tip retracts away from the surface. Stage 5 shows adhesion forces causing a hysteresis loop as the tip remains in contact. Finally, in stage 6, the tip detaches, returning to the initial no-interaction phase.

A constant peak force is applied to scan the surface of the sample and the deflection of the cantilever is measured. In stage 4, Dejaguin Muller Toporov (DMT) model was used to fit the curve between the maximum force and the maximum adhesion [95]. The reduced Young's modulus E_r is given by the following formula:

$$F = \frac{4E_r\sqrt{R(d-d_0)^3}}{3} + F_{adh} \tag{5}$$

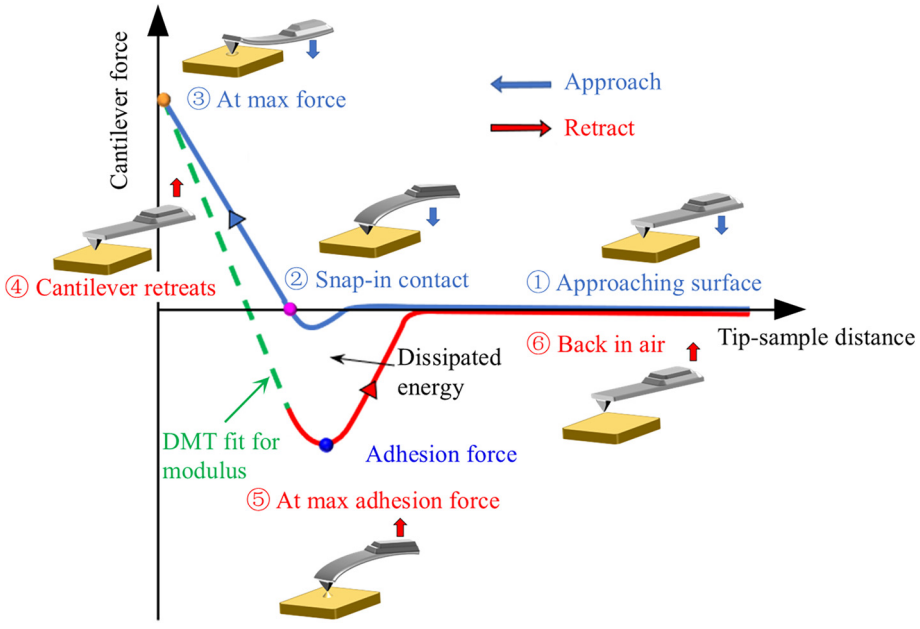


Fig. 7. The process of the force response between the AFM probe and the sample surface.

where F is the force acting on the probe tip; R the radius of the probe tip; $(d-d_0)$ the distance between the tip and the sample; and F_{adh} the adhesion force between the tip and the sample. The reduced Young's modulus E_r can be converted to the absolute Young's modulus E_{AFM} by the following formula:

$$E_r = \frac{1}{\frac{1-v_{AFM}^2}{E_{AFM}} + \frac{1-v_a^2}{E_a}} \quad (6)$$

where v_{AFM} is the Poisson's ratio of the sample; v_a the Poisson's ratio of the probe; and E_a the Young's modulus of the probe.

3.2.2. Advantages of AFM

AFM can achieve atomic or near-atomic resolution, allowing visualization of individual atoms or nanostructures on sample surface. It can operate in various environments, such as air, vacuum, or liquids, enabling studies under different conditions [96,97]. AFM probes a wide range of surface properties, including topography [98], magnetic fields [99], electrical properties [100], and mechanical properties [93]. It is widely used to study the nanoscale topography and morphology of surfaces, including geological samples, providing insights into mineral and rock formation processes and structural properties. AFM offers high-resolution (<2 nm) and 3D imaging of geological surfaces, characterizing features like roughness, cracks, and pores, which are essential for understanding tribological properties and weathering processes. It measures the nanomechanical properties of rocks and minerals by applying force and measuring the deformation to obtain local mechanical properties such as elastic modulus and hardness. AFM also measure the adhesive and frictional forces, providing information on the surface interaction and bonding mechanisms. When combined with techniques like AFM-IR or AFM-Raman, chemical information at the nanoscale can be determined [101,102].

3.2.3. Applications of AFM in planetary rocks and geological samples

In the laboratory studies of planetary rocks, AFM has been used to analyze the surface properties of lunar regolith, Martian meteorite fragments, and cosmic dust particles. Direct measurements of the mechanical properties of planetary rock samples are currently limited [73,103,104]. Table 2 lists previous AFM applications in studying the mechanical properties of planetary rocks and geological samples. Bones et al. [104] measured the mechanical properties of Jbilet Winselwan (JW) meteorite samples before and after heat treatment to examine the mechanical changes resulting from the pyrolysis of organic constituents during atmospheric entry. The findings suggest that while the residual meteoritic particles

Table 2
Overview of mechanical properties characterized by AFM.

Sample	Classification	Mineral phases	Reported properties	Values
Planetary rocks				
Jbilet Winselwan [104]	CM2 carbonaceous chondrites		Initial yield stress before and after heat treatment	Before 1.2–1.9 MPa After 1.9 MPa
HaH 346 [73]	L6 chondrite	Mineral interphase	Elastic modulus	16.92–23.24 GPa
BYX002 [47]	Lunar breccias clast		Elastic modulus	23.53 GPa (F=7.5 μN) 30.63 GPa (F=15 μN)
BYX005 [47]	Lunar breccias clast			4.20 GPa (F=7.5 μN) 5.65 GPa (F=15 μN)
JSC-1 [103]	Lunar simulants		Cohesive forces	Varies with humidity, grain size, and tip size
Earth samples				
Coal [111]		Organic matter Minerals	Elastic modulus	2 GPa 21 GPa
Shale [116]	Eagle Ford	Organic matter Clay Calcite Quartz, feldspar	Elastic modulus	9.8±1.1 GPa 22±2.0 GPa 50±9.2 GPa 5.5±10.1 GPa
Shale [115]	Upper Jurassic source rock	Pyrite Quartz Calcite Clay minerals Organic matter	Elastic modulus	>90 GPa 63±8 GPa 53±6 GPa 29±1 GPa 0–25 GPa
Organic matter [94]	Bakken Shale	Immature Oil window Thermally mature	Elastic modulus	1.46–9.69 GPa 1.32–5.94 GPa 4.99–22.34 GPa
Organic matter [122]	Cretaceous source rock	0.4 %VRo 0.82 % VRo (kerogen) (bitumen) 1.25 %VRo(kerogen)(bitumen)	Elastic modulus	3.2–7.8 GPa 7.7–29.2 GPa 6.2–10.1 GPa 9.3–29.9 GPa 5.4–17.1 GPa
Chlorite [120]		Chlorite	Elastic modulus	14–43 GPa
Granite [93]			Elastic modulus	1113.43 MPa
Shale [93]			Elastic modulus	540.29 MPa
Limestone [93]			Elastic modulus	1126.71 MPa
Basalt [118]			Elastic modulus Hardness	29.85 GPa 6.19 GPa
Limestone [118]			Elastic modulus Hardness	49.79 GPa 0.29 GPa
Diabase [118]			Elastic modulus Hardness	29.43 GPa 9.83 GPa
Granite [118]			Elastic modulus Hardness	35.81 GPa 9.91 GPa

become more brittle after organic pyrolysis, they also become slightly harder. Jardine et al. [103] studied adhesive forces between JSC-1 simulant particles and AFM tips to better understand the asteroid regolith cohesion. Tang et al. [73] measured the Young's modulus of interfaces between different rock-forming minerals in the HaH 346 meteorite (Fig. 8), using simulations to evaluate macroscopic mechanical behavior. AFM helps study the fatigue and fracture behavior of planetary rocks [26,105]. This is important for understanding how rocks on other planets may degrade over time due to repeated thermal cycling, impacts, or other stresses.

Besides mechanical characterizations, AFM has been widely used to investigate the surface morphology, mineralogy, and potential biosignatures on extraterrestrial rock samples [106]. Ming and Morris [107] analyzed Martian dust and soil, revealing that the surface of Mars is covered with fine-grained, basaltic material enriched with sulfates, chlorides, and nanophase iron oxides. AFM studies of lunar samples returned by the Apollo 11 and Chang'e-5 mission, provided insights into the shape and distribution of lunar dust particles, which is crucial for understanding weathering processes and magnetic properties of lunar surface [47,108]. AFM has also analyzed the fine structure of meteorites surfaces, including potential biological traces on Martian rocks [15]. Combining AFM with Raman spectroscopy allows researchers to map geological contexts and identify carbon deposits, detecting subtle traces of life such as nanofossils and organic filaments [14]. AFM technology is also being developed for space missions. In ESA's MIDAS-Rosetta mission, AFM conducted microtextural and statistical analysis of comet dust particles [109]. In NASA's Phoenix mission, AFM was used to obtain in-situ nanometer-scale images of Martian dust, making the first use of AFM for imaging the Martian surface [92]. Ongoing research aims to improve AFM technologies, such as instrument operation algorithms [91], micromachined silicon substrates [110], and data interpretation, for better in-situ investigations in planetary exploration activities.

Compared with planetary rocks, AFM technology is more widely reported in the mechanical study of energy geological materials, such as coal [111,112], shale [113–116], sandstone [117], limestone [93,118], granite [93,119], and clay minerals [120]. Tang's team use AFM to measure the elastic modulus of interfaces between different minerals within rocks, which is important for understanding the overall mechanical rock integrity [119]. Many studies employ AFM to directly measure the Young's modulus

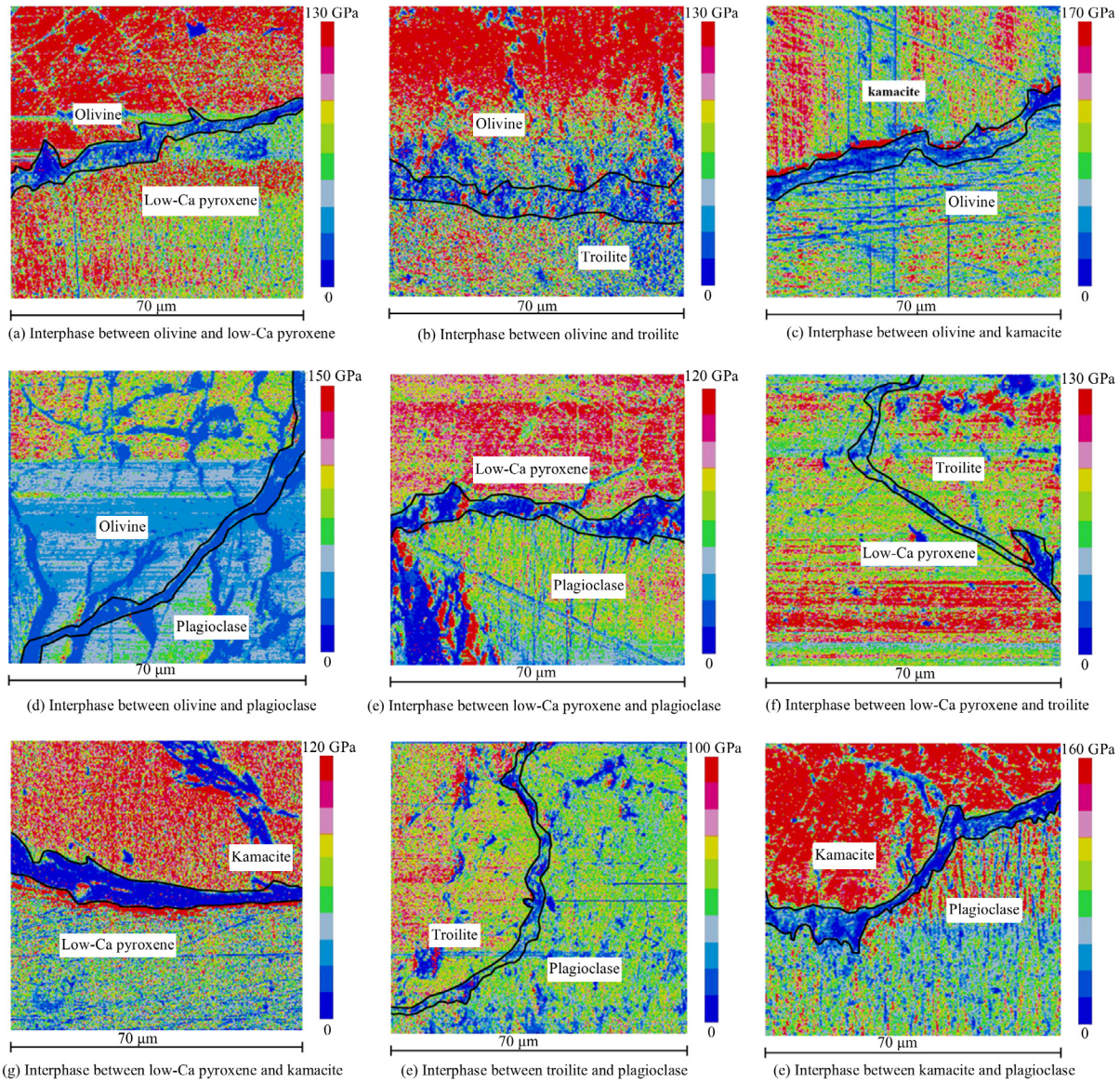


Fig. 8. Young's modulus of interfaces between rock-forming minerals in the HaH 346 meteorite (adapted from [73]).

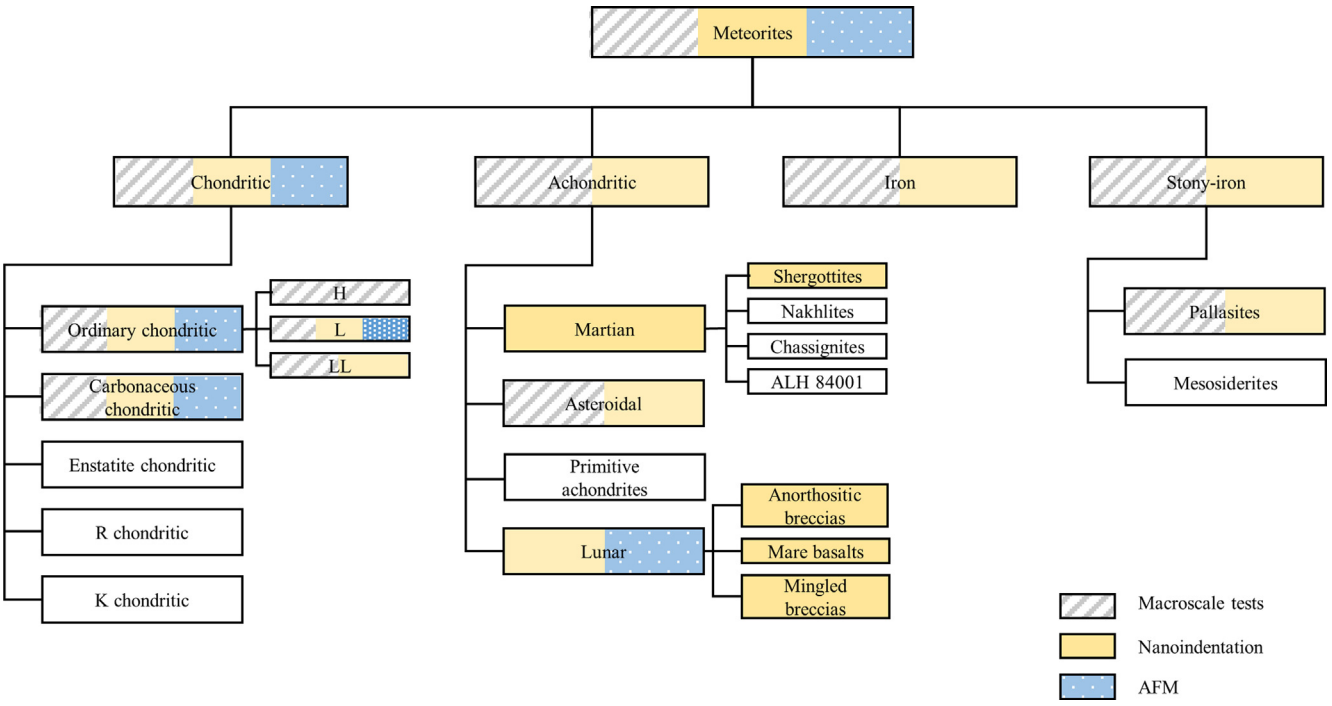


Fig. 9. Mechanical properties of meteorites corresponding to their types and measurement methods.

and hardness of organic matter in shales, particularly its Young's modulus distribution under different maturation levels and depositional environments, providing valuable implications for shale gas development, hydraulic fracturing optimization, and CO_2 geological storage. Eichmann et al. [114] used AFM-Raman to explore the relationship between source rock maturity and the mechanical properties of organic matter to predict the production potential of reservoirs. Wang et al. [101] investigated the nanoscale geochemical heterogeneity of organic matter in thermally-mature shales using AFM-IR. Besides measuring elastic modulus and hardness, AFM accesses adhesion force, rock stability mineral interactions, and wettability, essential for oil and gas processes [111,121]. AFM is also a powerful tool for observing in-situ dynamic processes on mineral surfaces, such as dissolution, precipitation, and weathering, recording by time-lapse nanoscale imaging in an AFM Fluid Cell attachment.

3.3. Analysis of the available data

3.3.1. Mechanical measured meteorite category

We compiled the mechanical properties of various meteorite types available in the literature, as shown in Fig. 9. It should be noted that meteorite classification can be very complex. Since this is not the focus of our review, we have simplified the classification of meteorites to present more intuitive results. The mechanical data of meteorites from macroscopic mechanical tests, nanoindentation, and AFM are summarized. As shown in Fig. 9, the available mechanical data in the literature cover ordinary chondrites, carbonaceous chondrites, achondrites, iron meteorites, and stony-iron meteorites, encompassing all major categories. In more detailed subcategories, relatively common meteorites, such as ordinary chondrites, have both macroscale and microscale mechanical test data. Additionally, we have summarized the amount of available mechanical data for different types of meteorites. Macroscale mechanical data are sourced from a previous review [24], as shown in Fig. 10. It is observed that current macroscale mechanical testing of meteorites primarily focuses on L-type meteorites and iron meteorites, with mechanical data for other types of meteorites being sparse. Microscale mechanical data are also limited across various meteorite types. The macroscale mechanical data is more than the available microscale data for chondrites, carbonaceous chondrites, and iron meteorites. For pristine samples such as Martian and lunar meteorites, micromechanical testing has played a significant role in evaluating their mechanical properties, given the absence of macroscopic mechanical test results.

Due to the limited availability of meteorite samples, the current micromechanical data are still sparse, limiting our understanding of mineral properties, microstructure, and formation processes in meteorites. One promising approach is to use machine learning and big data to predict the mechanical properties of meteorites, similar to the prediction of protein structures. However, this approach requires a substantial amount of mechanical data to support the operation of large models. Currently, the available data-

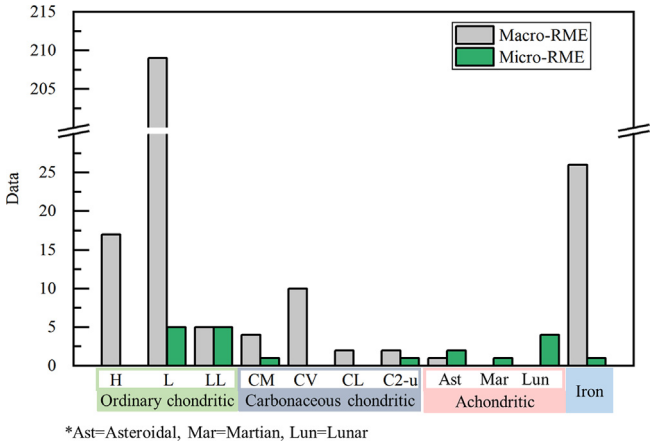


Fig. 10. Number of available cases in the literatures.

Table 3
Summary of probed mechanical properties of rock-forming minerals.

Mineral phase	Chemical formulas	Elastic modulus (GPa)	Hardness (GPa)
Augite	$(Ca, Mg, Fe, Al)(Si, Al)_2O_6$	94–160 [90]	
Calcite	$CaCO_3$	63–74 [16]	2.4–2.8 [16]
Clinopyroxene	$(Ca, Mg, Fe)Si_2O_6$	137.55±7.44 [82]	
Chromite	$FeCr_2O_4$	153.1±9.4 [31], 131±3 [12], 147 [16]	16.7±0.9 [31],15.9±1.3 [12], 17.3 [16]
Chlorapatite	$Ca_5(PO_4)_3Cl$	42.67±2.17 [82]	
Plagioclase feldspar	$CaAl_2Si_2O_8$ $NaAlSi_3O_8$	59.4–86.6 [90], 70.74 [70], 79–99 [16], 45.86±12.95 [73], 85±5 [72], 74.13±0.93 [82], 74 [80]	7.4–10.6 [16], 6.9 [80]
Iron-nickel(Fe-Ni alloy)	(Fe, Ni)	178±6 [71], 87.24 [70], 156±5 [72]	
Ilmenite	$FeTiO_3$	91±13 [80], 103–127.85 [47], 134 [16]	8.4±0.6 [80], 7.43–9.91 [47], 9.5 [16]
Kamacite	(Fe, Ni) with typically 5%–10% Ni	211.3±9.4 [31], 127±16 [12], 241 [33] 150.96±7.54 [73], 116.73 [29]	3.3±0.1 [31], 3.58±0.24 [12], 3.55 [33], 2.555 [29]
Merrillite	$Ca_3Ti_2Si_3O_{12}$	40.28±11.22 [73]	
Maskelynite	Amorphous phase	94.94±4.31 [82], 86±34 [80]	7.2±2.8 [80]
Magnetite	Fe_3O_4	103.90±3.32 [82]	
Olivine	$(Mg, Fe)_2SiO_4$ Forsterite: Mg_2SiO_4 Fayalite: Fe_2SiO_4	202.7±10.2 [31], 136±5 [12], 116.73 [70], 124.19±16.69 [73], 127±5 [72], 110±32 [80], 119.06–142.04 [47], 113–174 [16]	16.4±0.8 [31], 13.6±0.9 [12], 7.6±2.2 [80], 10.12–12.00 [47], 8.8–12.5 [16]
Pyroxene	$(Ca, Mg, Fe)Si_2O_6$ Augite: $(Ca, Na)(Mg, Fe, Al, Ti)(Si, Al)_2O_6$ Enstatite: $MgSiO_3$ Diopside: $CaMgSi_2O_6$	175.9±10.8 [31] 95±3 [71], 122±11 [12], 101.77 [70], 106.63±13.78 [73], 123±5 [72], 91±12 [80], 44.77–71.35 [47], 134–162 [16], 94–160 [90]	12.7±1.0 [31] 11.9±2.2 [12], 7.1±1.3 [80], 6.12–9.73 [47], 9.9–14.3 [16]
Plessite	Mixture of kamacite and taenite	219 [33], 185.8 [29]	4.22 [33], 2.879 [29]
Schreibersite	$(Fe, Ni)_3P$	190.8 [29]	11.03 [29]
Spinel	$(Mg, Fe, Zn, Mn)(Al, Cr, Fe)_2O_4$	116–154 [16]	9.6–10.6 [16]
Troilite	FeS	108.0±3.4 [31], 71±8 [12], 93.36±6.07 [73], 147 [16]	3.9±0.2 [31], 5.1±0.8 [12], 12.7 [16]
Taenite	(Fe, Ni) with typically 20%–50% Ni	82±6 [12], 65 [33]	3.05±0.29 [12], 3.98 [33]
Tetrataenite	FeNi (ordered structure of Fe and Ni)	237 [33], 152.6 [29]	8.43 [33], 2.390 [29]

bases are insufficient and do not cover all types of meteorites. Therefore, future efforts should focus on establishing a comprehensive database of meteorite mechanical parameters. This database should correlate mechanical data with meteorite types, mineral composition, and temperature or impact history. By training with big data, it may become possible to predict mechanical properties from the mineralogical composition or images of meteorites. Future research should aim to fill these gaps, particularly for meteorites lacking mechanical property data.

3.3.2. Micromechanical properties of the minerals in meteorites

The general chemical compositions of minerals found on extraterrestrial surfaces, such as the Moon, Mars, and asteroids, reflect the unique geological histories and processes of these celestial bodies. Lunar minerals are primarily associated with anorthositic regolith breccias and mare basalts. Anorthositic regolith breccias are mainly composed of anorthite, a calcium-plagioclase mineral rich in aluminum and poor in iron. Mare basalts are dark crystalline igneous rocks formed from ancient lunar crust bombardment and basaltic magma floods, composed mainly of iron-rich pyroxene, olivine, ilmenite, and iron-poor plagioclase. Martian meteorites, grouped into the SNC types and ALH 84001, show signs of shock metamorphism and are composed of minerals like pigeonite, augite, maskelynite, iron-rich olivine, and the characteristic component iddingsite for Nakhlite. Chondritic meteorites commonly contain olivine, pyroxenes, iron-nickel minerals, and accessory minerals like troilite, iron oxide and plagioclase feldspar. Achondrites from the asteroid belt can include olivine, orthopyroxene, and various metal phases, reflecting the high-temperature processes and differentiation during their early formation.

A particularly interesting application of nanoindentation and AFM is mechanical phase mapping, which visualizes and characterizes different mineral phases within a meteorite. By combining with chemical analysis techniques such as EDS, it links the mechanical properties with meteorite's mineralogy. Measured meteorite rock-forming minerals including chromite [12,16,31], chlorapatite [82], calcite [16], plagioclase feldspar [16,70,72–74,80,82,90], iron-nickel [70,71], ilmenite [16,47,80], kamacite [12,29,31,33,73], merrillite [73], maskelynite [80,82], magnetite [82], olivine [12,16,31,47,70,72–74,80], plessite [29,33], pyroxene [12,16,31,47,70–74,80,82,90], schreibersite [29], spinel [16], troilite [12,16,31,73,74], taenite [12,33], and tetrataenite [29,33]. The obtained mechanical properties, mainly Young's modulus and hardness, are summarized in Table 3.

Fig. 11 illustrates the micromechanical properties of the major minerals in various types of meteorites. It can be observed that different mechanical properties across different meteorites, with a wide range of modulus values. Specifically, the modulus of olivine,

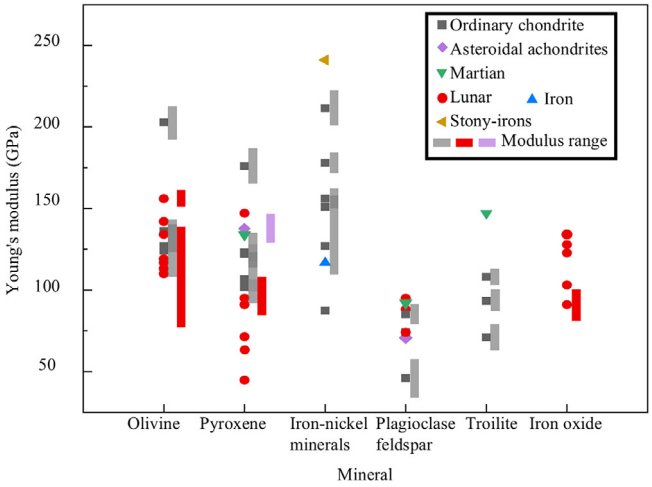


Fig. 11. Young's modulus of meteorites corresponding to their types using micro-RMEs.

pyroxene, and iron-nickel minerals in ordinary chondrites is generally higher than in lunar meteorites. This variation can be attributed to several factors. The formation environment of a mineral, including temperature, pressure, and cooling rate, significantly impacts its mechanical properties. Throughout their history, meteorites may undergo various stress events, such as impacts and deformations, which can introduce micro-cracks, alter dislocation density, and create other features within the minerals, thereby affecting their mechanical properties. Additionally, the same mineral may contain different trace elements or impurities, which can alter its mechanical properties. Meteorites that fall to Earth may experience weathering and other surface processes, which can further modify the surface conditions and internal structure of the minerals, thus affecting their mechanical properties.

3.3.3. Factors influence the data

It has been confirmed that testing parameters, such as loading rate, maximum load (or depth), and specimen preparation, significantly influence the accuracy and reliability of measurement results in micro-RMEs [123–125]. However, a uniform standard for loading rate cannot be provided because its effect depends on the material characteristics and the equipment. With technological advancements, new fast modes, such as XPM accelerated mapping mode, have emerged, increasing measurement speed by 500-fold [66]. It is recommended to select the appropriate loading rate based on the functions and instructions of the experimental instrument and it is best to do sensitivity analysis to determine the appropriate load.

Previous investigations emphasize the importance of selecting the appropriate load to accurately evaluate the mechanical properties of meteorites during nanoindentation testing [33,47,74]. Tanbakouei et al. [74] observed significant variations in contact stiffness under loads of 5 and 10 mN, while hardness showed no significant difference. Brusnitsina et al. [33] found that the hardness values obtained with a 1 mN load were slightly higher than those obtained with 7 mN, likely due to the scale factor effect at low loads. Nie et al. [47] demonstrated that maximum penetration depth, creep displacement, and contact stiffness increase with maximum load, affecting the Young's modulus and hardness of lunar regolith particles.

Two terms relevant to the impacts of load or depth on probing results are the indentation size effect (ISE) and substrate effect (SE). ISE is a phenomenon where the measured mechanical properties of materials decrease with increasing indentation depth due to the adjustment of internal material structure. SE occurs when the mechanical properties measured at larger indentation depths are influenced by the underlying substrate phases, leading to inaccurate assessments. Yang et al. [126] suggested an indentation depth range of 200 nm to 1/50 of the characteristic grain size for rocks to minimize the ISE and SE.

There is a wide of selection of tips for nanoindentation and AFM tests. The appropriate indenter should be chosen based on the material's characteristics and testing purpose. Common nanoindentation indenters include Berkovich, Vickers, and Cube Corner, each with unique geometrical characteristics and applications. Different indenter shapes affect the ratio of contact area to penetration depth, influencing the interpretation of test results. A brief overview of commonly used indenters is listed in Table 4. Berkovich indenters, with a three-sided pyramid tip, produce a larger contact area, minimizing crack formation for brittle materials. Vickers indenters have a four-sided pyramid with larger face angles, while Cube-corner indenters feature a sharp three-sided pyramid tip. These two sharper tips are suitable for testing fracture toughness by reducing cracking thresholds. AFM also offers a variety of indenters made from different materials, which will not be detailed here. Generally, it is necessary to select the indenter that best matches the sample being tested to reduce measurement errors.

Surface roughness of samples can significantly impact the results of nanoindentation and AFM tests. Since nanoindentation measures the contact area indirectly through penetration depth, natural surface roughness can cause errors in determining the contact area between the indenter and the specimen, affecting hardness and elastic modulus calculations. To reduce surface roughness, specimens can be polished using mechanical polishing or ion beam milling to increase surface flatness. Mechanical polishing involves progressively finer abrasives, such as abrasive paper (usually up to P5000), diamond suspensions, or silicon carbide, applied to a polishing pad or cloth, typically using a rotating polishing machine. Alternatively, ion beam milling offers a more controlled and precise approach, especially for delicate or nanoscale samples. This process can be performed either wet or dry, with dry polishing recommended for meteorites to prevent rapid oxidation of iron or iron-bearing minerals. Previous studies recommended that for homogeneous materials like metals, the indentation depth should be 20 times larger than the roughness average, while for porous media, it should be three times larger than the RMS roughness [127,128]. Unfortunately, few studies on meteorites disclose the roughness values of their samples, making it challenging to evaluate the reliability of nanoindentation data. Given the significant influence of surface roughness on results, future work should ensure roughness meets requirements and report these values in publications.

4. Upscaling of microscale mechanical properties

Since the first measurements of the microscopic strength of meteorites, there has been the inevitable problem of how to relate these measurements to the strength of the parent body (asteroid or small meteoroid) [24]. Scaling from small to large is a critical aspect. Microscale mechanical rock measurements of meteorites provide mechanical information about the local mineral phases with the meteorites and further cross-scale estimation is necessary for space engineering applications. This section summarizes methods widely used for upscaling the microscale mechanical properties of composite materials, mainly the methods that have been widely used in terrestrial rocks. These methods can be applied to meteorites in future works. We also list existing applications of these method on planetary rocks or simulants to assist researchers in the field.

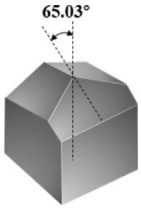
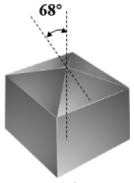
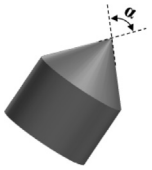
4.1. Homogenization schemes

Over the past few decades, homogenization methods have been developed to evaluate the overall elastic properties of heterogeneous materials and their dependence on compositive properties and material microstructure, based on Eshelby equivalent inclusion theory [129]. These methods are particularly useful for understanding the elastic properties of porous geomaterials, which are challenging to measure directly due to their heterogeneity and the difficulty in retrieving good-quality core samples. We introduce the common homogenization schemes from literatures used in upscaling nanoindentation results for rocks.

4.1.1. Mori-Tanaka (M-T) scheme

Mori and Tanaka [130] proposed a method for calculating average internal stress and elastic energy in the matrix of materials containing inclusions with transformation strain. They considered a hypothetical scenario where inclusions are uniformly distributed and have the same transformation strain. Using Eshelby's Green

Table 4
Shapes of different indenters for nanoindentation.

Indenter	Geometry	Descriptions
Spherical		• The stress distribution under the indenter is relatively uniform due to its geometric shape. • Suitable for measuring soft materials and replicating contact damage in service conditions.
Berkovich		• The edges of the pyramid meet at a single point, which is advantageous for precision in small-scale testing. • Suitable for hard materials, thin films, scratch testing, and wear testing.
Cube-corner		• With a semi-angle of 35.3°, it creates a larger stress concentration at the contact point. • Suitable for fracture toughness, scratch testing, thin films, and wear testing.
Vickers		• A four-sided pyramidal indenter with a face angle of 68°. • Suitable for hard materials, thin films, scratch testing, and wear testing.
Knoop		• A four-sided pyramidal indenter with two different face angles. • The unequal diagonal lengths of the residual impression are useful for determining the size/depth of the indentation. • Cube-corner shape. Useful for studying the anisotropy of the material surface.
Conical		• It is characterized by its cone angle and the radius of the tip curvature. • Its simple shape makes it easy to model. • Suitable for scratch testing and wear testing.

function and several assumptions, expressions for the average internal stress in the matrix were derived [129,130]. This stress is uniform and independent of the position and size of the calculation region. The Mori-Tanaka (M-T) scheme considers interaction among inclusions and the impact of the material's free boundary on average internal stress and elastic energy. The M-T method is particularly useful for materials with varying mechanical properties and isotropic characteristics at the microscale. It calculates the equivalent volume modulus K_{hom} and equivalent shear modulus G_{hom}, considering the porosity and the mechanical properties of the low-, medium-, and high-strength phases:

$$K_{\mathrm{hom}}^{\mathrm{M-T}} = \frac{\sum f_i \frac{k_i}{3k_i+4\mu_{\mathrm{low}}}}{\sum \frac{f_i}{3k_i+4\mu_{\mathrm{low}}}} (i = 0, 1, 2) \tag{7}$$

$$G_{\mathrm{hom}}^{\mathrm{M-T}} = \frac{\sum \frac{f_i\mu_i}{\mu_{\mathrm{low}}\left(9k_{\mathrm{low}}+8\mu_{\mathrm{low}}\right)+6\mu_i\left(k_{\mathrm{low}}+2\mu_{\mathrm{low}}\right)}}{\sum \frac{f_i}{\mu_{\mathrm{low}}\left(9k_{\mathrm{low}}+8\mu_{\mathrm{low}}\right)+6\mu_i\left(k_{\mathrm{low}}+2\mu_{\mathrm{low}}\right)}} \tag{8}$$

$$k_i = \frac{E_i}{3(1-2v_i)} \tag{9}$$

$$\mu_i = \frac{E_i}{2(1+2v_i)} \tag{10}$$

where k_i is the volume modulus of minerals; μ_i the shear modulus of minerals; E_i and v_i the elastic modulus and Poisson's ratio of different minerals; and f_i the volume fractions. i=0 represents the low-strength phase; i=1 the medium-strength phase; and i=2 the high-strength phase. To consider the impact of micropores in low-strength minerals, the equivalent volume modulus k_{low} and the equivalent shear modulus μ_{low} of low-strength minerals are calculated as follows (ϕ is the porosity):

$$k_{\mathrm{low}} = \frac{4(1-\phi)k_2\mu_2}{4\mu_2+3\phi k_2} \tag{11}$$

$$\mu_{\mathrm{low}} = \frac{(1-\phi)\mu_2}{1+6\phi\frac{k_2+2\mu_2}{9k_2+8\mu_2}} \tag{12}$$

The homogenized elastic modulus E_{hom} is then determined by:

$$E_{hom} = \frac{9K_{hom}G_{hom}}{3K_{hom} + G_{hom}} \tag{13}$$

The M-T method has been widely used to estimate the macroscale elastic modulus of rocks from nanoindentation test. It has been applied on various Earth rocks such as shales [67,113,131,132], granite [133], claystone [134], sandstone [125], mudstone [135], carbonate rocks [136], marble [137]. For asteroid meteorites, Tang et al. [70,73] utilized M−T method to calculate the macroscale elastic modulus of the NWA13618 and HaH 346 meteorite. They first obtained the elastic moduli and volume fractions of the main minerals (olivine, pyroxene, Fe-Ni metal, and feldspar) through nanoindentation testing and Gaussian mixture model. These microscale mechanical parameters were then upscaled to predict the mechanical properties of the L4 parent asteroid using the Mori-Tanaka homogenization

4.1.2. Self-consistent scheme

The self-consistent scheme (SCS) is a method for estimating the macroscale elastic moduli of two-phase composite materials. Proposed by Hershey [138], Kröner [139] and later Hill [140], it accounts for stress and strain inhomogeneity and applies to materials with arbitrary shape and concentration, requiring a matrix and effectively ellipsoidal inclusions. Initially used for crystal aggregates, the SCS method assigns typical crystal properties to the inclusion and macroscopic properties to the matrix, offering a direct analysis approach for composite materials. This method was later applied to geomaterials, deriving relationships between overall moduli and phase moduli through statistically homogeneous distributions of inclusions [113,125,141]. The calculation of SCS is as follows:

$$K_{hom} = \sum_{i=0} f_i \frac{k_i(3K_{hom} + 4G_{hom})}{3k_i + 4G_{hom}} \tag{14}$$

$$G_{hom} = \sum_{i=0} f_i \frac{5\mu_i G_{hom}(3K_{hom} + 4G_{hom})}{G_{hom}(9K_{hom} + 8G_{hom}) + 6\mu_i(K_{hom} + 2G_{hom})} \tag{15}$$

where K_{hom} and G_{hom} are the equivalent volume modulus and shear modulus; k_i the volume modulus of the i-th mineral phase; μ_i the shear modulus of the i-th mineral phase; E_i the elastic modulus of mineral particles; and v_i the Poisson's ratio of different minerals. As can be seen from Eqs. (14) and (15), the analytical formula of parameters cannot be directly obtained and usually be solved by numerical iteration.

The M-T scheme considers matrix perturbations due to the presence of all inclusions, whereas the SCS includes interaction among inclusions without treating any phase as a matrix. This difference may cause the SCS to inaccurately predict the overall elastic properties of composites [142]. The authors did not find literature involved in the application of SCS to the multi-scale mechanical properties of planetary rocks. However, Hazeli et al. [72] used Mori-Tanaka model to quantify microscale stress distribution caused by diurnal temperature variations and understand the impact of thermal fatigue on rock microstructure.

4.1.3. Voigt-Reuss-Hill average method

The Voigt-Reuss-Hill average method combines the Voigt (upper) [143] and Reuss (lower) [144] bounds to evaluate the effective elastic modulus of composite materials. The Voigt model assumes a series configuration of individual constituents, with the elastic modulus being the average of the moduli of the individual constituents, weighted by their volumetric fractions (γ_i):

$$E_v = \sum_{i=1}^{n} \gamma_i E_i \tag{16}$$

The Reuss model stipulates a parallel configuration of individual constituents and calculates the effective modulus as the harmonic mean of the individual moduli:

$$\frac{1}{E_R} = \sum_{i=1}^{n} \frac{\gamma_i}{E_i} \tag{17}$$

Hill's average, which is the simple arithmetic average of the Voigt and Reuss bounds, is used to estimate the effective elastic modulus of a composite material [145]:

$$E_H = \frac{1}{2}(E_v + E_R) \tag{18}$$

The Voigt-Reuss-Hill average method has been applied to shale [141], single crystal and minerals [146,147], sandstone [148], and siltstone [149] to estimate macroscale elastic modulus, Poisson's ratio, brittleness, wave velocity [150] of rocks. Tang et al. [73] employ Voigt-Reuss-Hill method to determining the macroscale Young's modulus of HaH 346 asteroid meteorite based on the micro-RMEs results of rock-forming minerals. The results indicated that Voigt-Reuss-Hill method can lead to an error of 85.57%, while M−T method show an error of 73.53%. The study suggests that the accurate grain-based model (AGBM) is the most accurate method, with an error of 13.21% and is therefore used to estimate the macroscale Young's modulus of HaH 346 meteorites. A dedicated introduction on AGBM is presented in section 4.2.1.

4.1.4. Dilute scheme

The dilute method assumes that all mineral materials are continuous, uniform, and isotropic linear elastomers, with mineral crystals randomly distributed and representative volume units isotropic. Since the interaction between each mineral crystal is not considered, the effective mechanical properties of the representative volume units can be regarded as a simple superposition of the inclusion of each mineral crystal. The dilute method has been applied to Earth rocks [134,151], but not to planetary rocks to date. The equivalent volume modulus and shear modulus can be determined as follows:

$$K_M^{dil} = k_0 + \sum_{i=1} f_i \frac{(k_i - k_0)(3k_0 + 4\mu_0)}{3k_i + 4\mu_0}(i = 0, 1, 2) \tag{19}$$

$$G_M^{dil} = \mu_0 + \sum_{i=1} f_i \frac{5\mu_0(\mu_i - \mu_0)(3k_0 + 4\mu_0)}{\mu_0(9k_0 + 8\mu_0) + 6\mu_i(k_0 + 2\mu_0)} \tag{20}$$

Among the methods discussed, the M−T method accounts for the interactions between inclusions to determine the elastic properties of the clay matrix. Previous studies indicate that the M−T model can accurately replicate variations in macroscopic elastic modulus across different geomechanical zones, whereas the SCS may overestimate material stiffness, especially at high concentrations of inclusions [152]. The Voigt-Reuss-Hill method provides a balance between upper and lower bounds, offering a straightforward and quick estimation. The dilute scheme, assuming no interactions between phases, may be inaccurate for materials with significant phase interactions [125]. Additionally, other homoge-

nization methods applied to rocks include the generalized means method [133], and differential effective medium (DEM) theory [125,153].

4.1.5. The representative volume element (RVE) homogenization method

The RVE homogenization technique is a powerful method in materials science and engineering, particularly for analyzing composite materials. The fundamental principle of this technique is to capture the microstructure of a material within a small, representative volume, which can then predict the effective or homogenized properties of the material at a macroscale level. Determining the appropriate RVE size is a critical step in the homogenization process for heterogeneous materials. It must capture essential microstructural features (grains, inclusions, voids, etc.) while remaining computationally feasible. The RVE should contain enough different phases present in the composite and be large enough to capture the relevant microstructural features but small enough to avoid scale effects associated with the material's macroscale behavior.

Usually, the RVE is determined by conducting a convergence study where the effective properties (e.g., elastic modulus, permeability, conductivity) are calculated for a series of progressively larger volumes of the material. This involves increasing the RVE size and observing changes in the calculated effective properties. Statistical methods, such as the concept of integral range, analyze the variability of the apparent properties as a function of the volume. Numerical simulations, like finite element analysis, can compute the apparent properties of different-sized volumes and identify trends and biases. The minimal size that results in convergence of the effective properties can be defined as the RVE, indicating that this size is sufficient to capture the material's overall microstructural characteristics, leading to the determination of its macroscale representative properties.

Gholami et al. [154] examined the hybrid microwave sintering of a lunar soil simulant using the RVE homogenization technique to determine the effective stiffness (or elastic modulus) of the sintered product. The authors used image analysis to determine the volumetric fractions of differently colored contrasts in the microstructure of the sintered specimens, corresponding to different mineral phases with varying elastic moduli. These phases include anorthite, augite, and iron (titanium) oxide, each with a distinct elastic modulus as determined by nanoindentation. The equation for the effective elastic modulus based on the RVE homogenization technique is as follows:

$$E = \frac{\sum_{i=1}^{n} E_i V_i}{\sum_{i=1}^{n} V_i} \tag{21}$$

where E_i is the elastic modulus of the i-th phase; V_i the volumetric fraction of the i-th phase; and n the total number of distinct phases present in the material. By applying this equation, the researchers computed the effective elastic modulus of the sintered lunar soil simulant, considering the microstructural characteristics and the mechanical properties of its constituent phases. This approach is essentially analogous to the Voigt model.

4.2. Numerical simulations: Microscale to macroscale modeling

Numerical techniques are efficient tools for understanding the complex multiscale mechanical behavior of rocks. Microscale to macroscale modeling helps translate microscale observations into macroscale properties, predicting the behavior of rocks under various conditions. This involves upscaling techniques that integrate the properties of individual grains and their interactions to estimate the macroscale properties of the bulk rock mass.

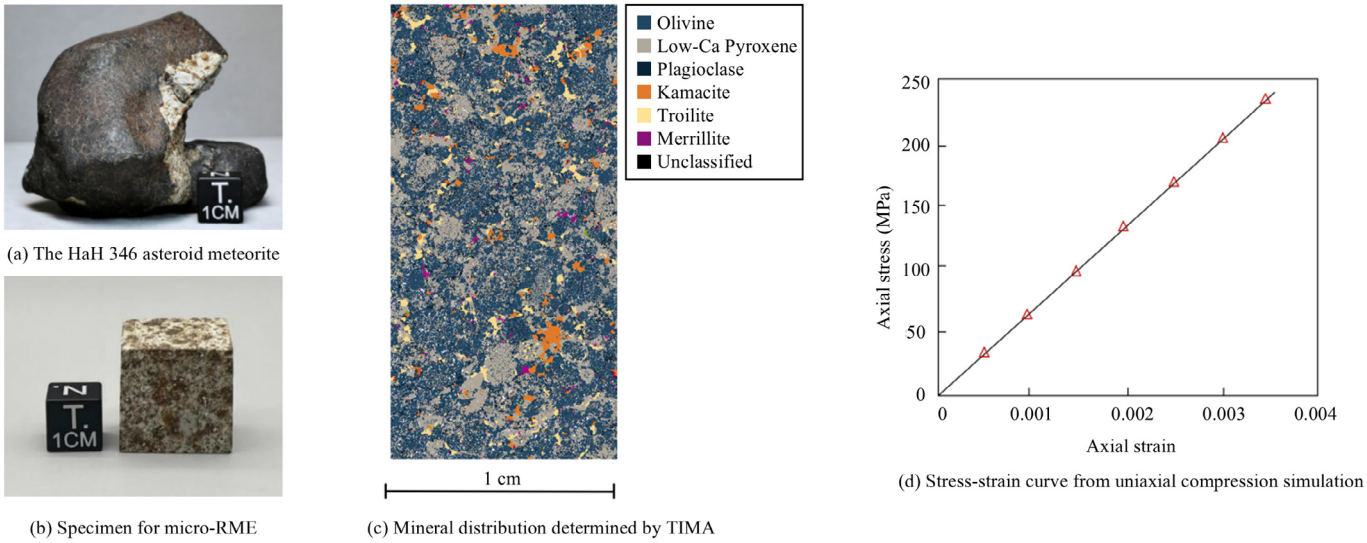
The grain-scale perspective is particularly important because it allows researchers to account for the heterogeneous nature of rocks, including both compositional (e.g., elemental and mineralogical) and microstructural (e.g., particle size, shape, packing, and pores and their size distributions) non-homogeneity. Numerical techniques have been employed to model these microstructural features and their impact on the macroscale properties of bulk rocks. We introduce the modeling methods based on mineral grains and their applications to studies of planetary rocks.

4.2.1. Accurate grain-based modeling (AGBM)

Grain-based modeling (GBM) is a numerical simulation technique that operates at the scale of individual mineral grains in rocks. It utilizes Voronoi tessellation to generate polygonal grain topologies, simulating the mechanical behavior of rock grains and revealing the complex micro-to-macro response mechanisms of rocks from the perspective of individual grains. Abdelaziz et al. [155] employed Voronoi mesh to simulate the crystal structure of rock minerals, revealing that the macroscale failure paths of rocks result from the combination of local stresses and microstructure. Peng et al. [156] utilized the GBM method to simulate different failure modes of granite. Molaro et al. [157] utilized GBM to simulate thermoelastic grain-scale stresses induced by diurnal temperature variations on various solar system bodies.

However, the Voronoi mesh generation method in traditional GBM typically simplifies the geometric shapes of rock-forming minerals based on statistical grain size distribution, failing to accurately depict the geometric distribution of mineral particles. Additionally, the mechanical parameters of minerals and mineral interfaces are unclearly defined. Properties that are difficult to measure experimentally are estimated through numerical calibration, adjusting model parameters until simulated macroscale behavior matches results from macro-RME tests, which are both imprecise and sampling requiring [119]. Moreover, it ignores interphase mechanical properties between different grains, which can significantly impact macroscopic rock behavior, including strength, stiffness, and failure modes.

To overcome traditional GBM limitations, Tang et al. [119] proposed accurate grain-based modeling (AGBM) by incorporating detailed microstructural and microscale mechanical property data from micro-RMEs. The method has been used to determine the macroscale mechanical parameters of meteorites such as NWA13618 [70], HaH 346 (Fig. 12) [73], and NWA12564 Martian meteorites [82]. It was also used to study granite strength evolution, microcrack development under thermal loading [158] and weathering effects on granites' mineral grains and macroscale Young's modulus [159]. The flowchart of the method is illustrated in Fig. 13. TESCAN integrated mineral analyzer (TIMA) was adopted to acquire the mineral composition and the real microstructure of rock. With BSE and EDS probe, TIMA identifies mineral grains and creates a digital rock image with mineralogy and microstructure information. Subsequently, a 2D finite element model was reconstructed based on the digital image. The model accurately represents the rock structure in mineral fraction, mineral morphology, mineral sizes, occlusal relationship, and micro-voids. Micro-RMEs were used to determine the mechanical properties, with individual minerals measured via nanoindentation testing and the interphase Young's modulus determined by AFM. Combining AGBM with micro-RMEs results, numerical simulations were conducted to determine the macroscale Young's modulus and uniaxial compression strength of the rock. AGBM provides a tool for predicting rocks' macroscale mechanical behavior from microscales, especially in heterogeneous multiphase composite materials, currently only in 2D with plans for 3D models.



(a) The HaH 346 asteroid meteorite

(b) Specimen for micro-RME

(c) Mineral distribution determined by TIMA

(d) Stress-strain curve from uniaxial compression simulation

Fig. 12. Macroscale mechanical properties of HaH 346 asteroid meteorite determined using the AGBM (adapted from [73]).

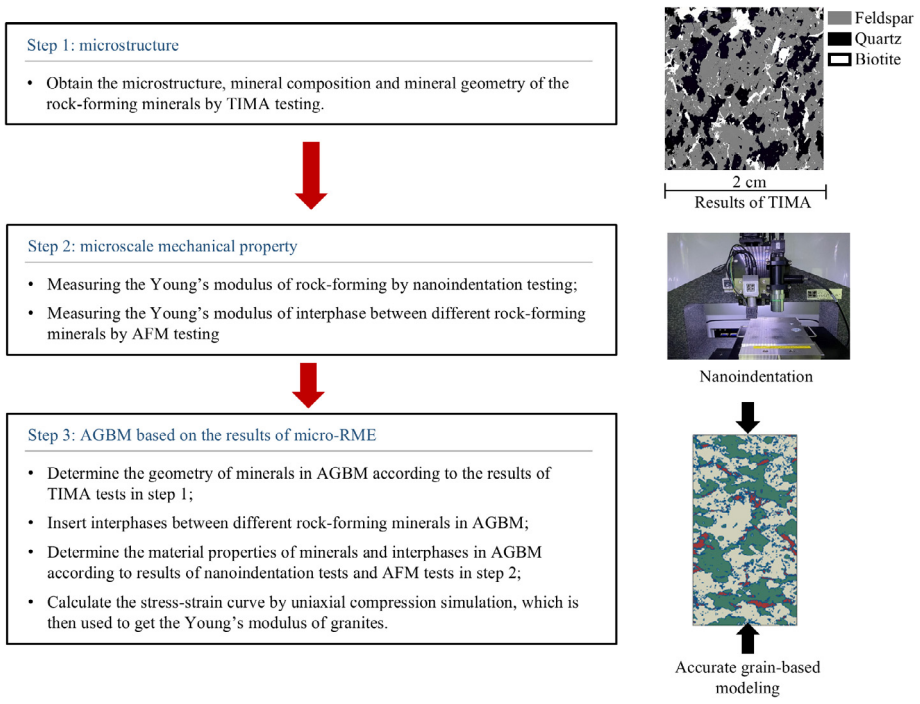


Fig. 13. Flowchart of estimating the macroscale mechanical properties of rock samples using the AGBM with micro-RMEs (adapted from [119]).

4.2.2. Three-dimensional (3D) numerical modeling

A significant challenge in simulating the micromechanical behavior of rocks is creating realistic 3D models of their complex microstructure. Techniques such as serial sectioning, XCT, electron tomography, and atom probe tomography (APT) can be used to reconstruct and visualize microstructural features like grains, pore spaces, and fractures. Hazeli et al. [72] used a 2D finite element model to study thermal fatigue and the microstructure of the GRO 85209 chondrite. XCT revealed the internal 3D microstructure, while local mechanical properties, including Young's modulus, shear modulus, Poisson's ratio, and coefficient of thermal expansion, were obtained using experiments. Results revealed the fatigue cracks propagation along the interfaces of individual minerals. Building on the Hazeli's work, Liang et al. [53] developed a 3D numerical model incorporating realistic asteroidal rock microstructure (Fig. 14). The study examined internal phase interactions, stress concentration along mineral interfaces, and stress development from mechanical loading, providing insights into damage initiation mechanisms through linear elastic response analysis.

The RVE of the digital rock model is crucial for obtaining reliable computational results. Models constructed from CT images include extensive pixels representing various material phases and complex microstructures. Balancing accuracy and computational cost make RVE size important. Previous studies summarized its influence on stress–strain responses [160]. RVE size depends on several factors,

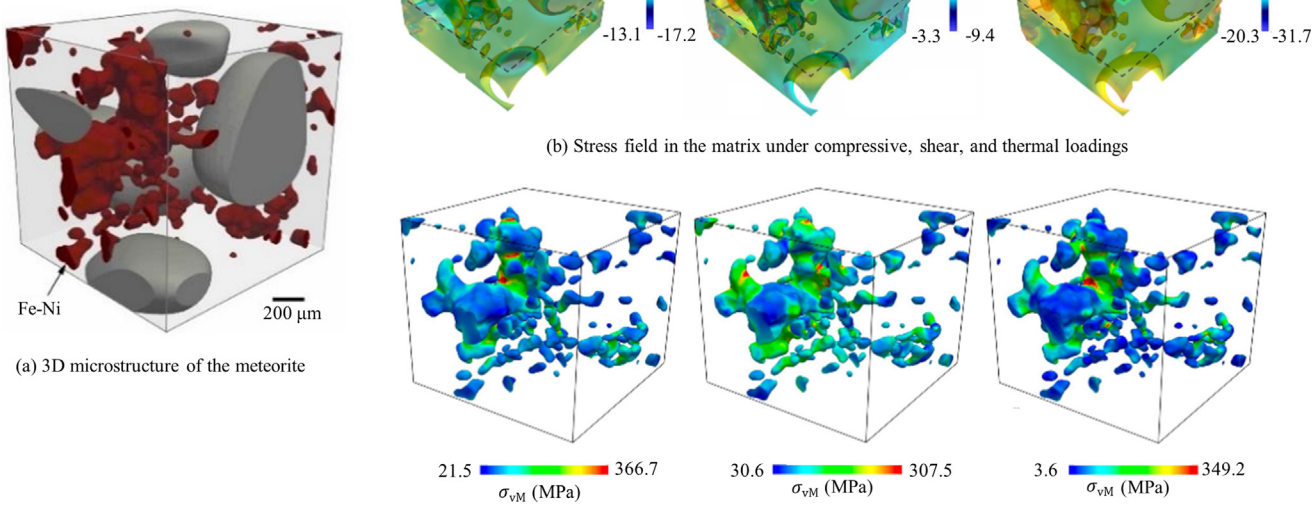


Fig. 14. 3D finite element analysis of a meteorite based on XCT images (adapted from [53]).

including the physical property of interest, the contrast in properties of the constituents, volume fractions, the desired precision in the estimation, and the number of microstructure realizations [161]. Finite element analysis software, such as Abaqus [61], OOF, OOF2 [72,157], were used to simulate stress–strain responses under various mechanical loading conditions.

Several challenges exist in constructing and analyzing 3D models of meteorites, including the complexity of microstructures, difficulties in image processing and segmentation, the need for high-quality finite element mesh generation, complexities in setting boundary conditions, limitations in computational resources, and the complexity of dealing with experimental data. Moreover, the current research on damage models for meteorites is limited, preventing a comprehensive understanding of damage mechanisms of meteorites. Future work needs to develop more sophisticated models that account for nonlinear material behaviors (such as damage growth and interface debonding) and strain rates (due to impact loading) to fully capture the mechanical failure processes of meteorites. Given the scarcity of asteroid samples and the difficulties associated with experimental testing, properly calibrating and validating these models with actual material properties is a challenging task that requires further research in this field. Despite these challenges, the development of 3D models is crucial for gaining a deeper understanding of the micromechanical behavior of meteorites, revealing damage mechanisms under thermal fatigue and mechanical loading, and has direct implications for understanding the surface evolution of asteroids, improving sampling methods, and planning future space missions.

5. Prospective applications closely related to meteorites' mechanical properties

5.1. Sample return missions

Sample return missions involve collecting real planetary materials from celestial bodies, such as asteroids, comets, or planets and bringing them back to the Earth for detailed analyses. These missions are crucial for understanding extraterrestrial geological structures and exploiting space resources, which helps the long-term support for the sustainable prosperity of the human society. Attempts have been made to obtain extraterrestrial samples from the Moon, Mars, and asteroids. These missions are complex and technologically demanding, as the success of coring depends on the coring device's performance and the distinct physical and mechanical properties of planetary soils and rocks. During the Apollo and Luna missions, lunar regolith experienced significant mechanical and thermal disturbances due to technical constraints, which led to the partial loss of the core samples' original compositions [162]. Thus, investigating the mechanical properties of planetary samples under extreme space environmental conditions is essential for designing and optimizing sampling equipment.

For landing and lander design, understanding the friction characteristics and shear strength of the planetary surface helps engineers predict the interaction between lander and the surface of the Moon, Mars, or other celestial bodies, leading to more stable landing devices. The choice of landing site also pays particular attention to the surface conditions [163]. Thus, obtaining as much physical and mechanical information as possible about the soils and rocks on the planet's surface is crucial. The strength data of meteorites can predict the mechanical response of materials on an asteroid's surface, planning safer and more efficient landing and sampling paths.

Designing mechanical systems for sample collection in complex and unknown environments of celestial bodies is a significant technical challenge. When conducting drilling and sampling activities, understanding the hardness, compressive strength, and interparticle adhesion force of meteorite samples is essential. This information aids in selecting appropriate drilling tools and methods to ensure effective penetration and rock sample collection while avoiding drill bit damage or sample degradation [164]. On the Earth, uniaxial compressive strength values are often used to characterize the drillability of rock types. However, there is limited

data on Martian surface strength [165]. At this point, microscale rock mechanics is an excellent choice to estimate the strength of rocks on Mars. Combining micromechanical test methods and scale-up techniques helps evaluate the drillability of planetary surfaces. Regolith simulants have been used in lab drilling tests to assess the mechanical and thermophysical status of sampling, including mechanics, motion, fatigue, and control [166]. The mechanical properties of the regolith are vital in all these processes. Meteorite strength data are used to assess risks in acquisition missions, such as sample breakage or tool damage.

Sample preservation and return are the final stages of sampling missions. Minimal changes in sample composition or morphology are required during collection, transit to Earth, entry into the atmosphere, surface impact and long-term storage [167]. Sample storage containers and processing procedures are designed to maintain the sample's integrity and original state based on meteorite's physical properties. Special packaging and transportation methods can reduce sample damage during the return to Earth, considering the meteorite's strength and vulnerability.

5.2. Planetary defense

Asteroids and comets intersecting Earth's orbit pose a significant risk of impact, potentially causing major disruption to the Earth and human society. Historical instances, like the Tunguska explosion [168], the Chelyabinsk event [169], and the hypothesized dinosaur extinction event [170], highlight the impact of asteroid collisions. These impact events have left a clear mark on the Earth's geological record [171]. While most meteorite falls are minor, events like the Chelyabinsk incident demonstrate their potential for widespread damages and injuries. Extensive research on asteroid impact hazards and mitigation strategies has led to significant progress in detecting and monitoring large Near-Earth Objects (NEOs) [13]. Various asteroid defense initiatives have been launched, such as Asteroid Redirection with Gravity Tractoring and Observation (ARGO), Asteroid Impact and Deflection Assessment (AIDA), NEOShield, and the B612 Foundation's Sentinel Mission. These initiatives employ diverse methods like gravity tractor [172], kinetic impactor [79], nuclear explosion [173], laser [174], and solar collector [175] to alter the orbits of potentially hazardous celestial bodies and prevent collisions with Earth.

The mechanical properties of meteorites are key parameters for formulating planetary defense strategies. In the ARGO mission, gravity traction technology is chosen to redirect the orbit of the secondary body in the 162,000 (1990 OS) binary Near-Earth Asteroid system, testing its efficacy in mitigating potential Earth impact hazards [172]. Implementing this technique requires the mechanical properties of asteroids to calculate gravitational forces and orbital dynamics accurately. In NASA's DART mission, the kinetic impact method was employed to alter the orbit of an asteroid in the Didymos binary system [176]. In the precedent investigation, meteoritic analogs like L and LL chondrites were used to estimate Didymos's density and physical properties, essential for predicting the DART mission's impact [177].

Other asteroid defense methods include nuclear, explosions, lasers, and solar collectors. Nuclear explosions involve detonating a device near the asteroids surface, while laser defense uses high-energy beams to evaporate or melt surface material, such as the DE-STAR (Directed Energy System for Targeting of Asteroids and Exploration) [178]. Solar collectors focus sunlight to increasing asteroid's temperature and causing material evaporation [174]. The asteroid's strength influences the outcomes of these methods, with cohesive asteroids resisting fragmentation from nuclear blasts and weaker ones potentially fragmenting under laser ablation or solar heating [174,175].

Meteorite mechanical properties also determine the energy released during impacts, with stronger meteorites causing more destructive impacts. Simulations based on these properties can study rock fragmentation, ejection production, and energy transfer, thereby enhancing the designs of planetary defense missions. Especially for the precious meteorite samples collected from asteroid, micro-RMEs offer valuable insights into asteroids composition and texture, aiding in accurate simulations and calculations before practical trials.

5.3. Extraterrestrial construction

Extraterrestrial construction involves the construction of buildings or habitats on celestial bodies like the Moon, Mars, or other planets within and beyond the solar system. It aims to create sustainable environments for human life, scientific research, exploration, and potential colonization. Due to high transportation costs, In-situ resource utilization (ISRU) is the most promising technique for sustainability and cost-effectiveness [179]. Consequently, the regolith, weathered soil on the planetary surfaces, is the primary available raw material for extraterrestrial construction. Thus, assessing the mechanical performance of regolith-based constructions is essential to ensure reliability in extreme extraterrestrial environments.

The mechanical performance of extraterrestrial construction materials based on regolith is of paramount importance. These materials face pressures, tensions, and bending moments during using. Thus, it is vital to determine compressive strength, flexural strength, and tensile strength through various tests. Considering the long-term usage, fatigue characteristics and impact resistance, particularly against micrometeoroid impacts, are also crucial. Due to the limited extraterrestrial regolith samples for destructive testing, regolith simulants are widely used [32,85,180]. Methods to consolidate regolith into load-bearing structures include cold sintering, laser sintering, geopolymerization, and crosslinking [90,181]. Additive manufacturing techniques like 3D printing are promising for generating construction parts [180,181]. Zhou et al. [19] provided a comprehensive investigation into the properties and characteristics of regolith materials. Currently, data on authentic regolith's physical, thermal, and mechanical properties are very limited, necessitating the development of non-destructive testing methods like microscale rock experiments. Space mining of NEOs is another potential resource for extraterrestrial construction (Fig. 15) [182]. Pilot missions like the Galileo and Hayabusa missions explored NEOs for potential mining operations [5,183]. Meteorites and dust particles from NEOs provide data for studying mechanical properties related to building and structural engineering, such as strength, Young's modulus, stress-strain response, thermal expansion, and creep properties [7,27,74]. These measured data points provide a glimpse into the composition and behavior of building materials mined from NEOs, critical for determining the mining method. For instance, NEOs with loose surface materials can be scraped using robotic attachments. Tunneling through cutters or drillers can be used to mine deeper surface and extract minerals.

6. Conclusions

In this paper, we review advances in the characterization of microscale rock mechanical properties and cross-scale estimation techniques for planetary rocks, which are crucial for deep space exploration missions. The following conclusions can be drawn:

(1) Non-destructive micro-RMEs on planetary rocks have made significant strides, offering profound insights into their

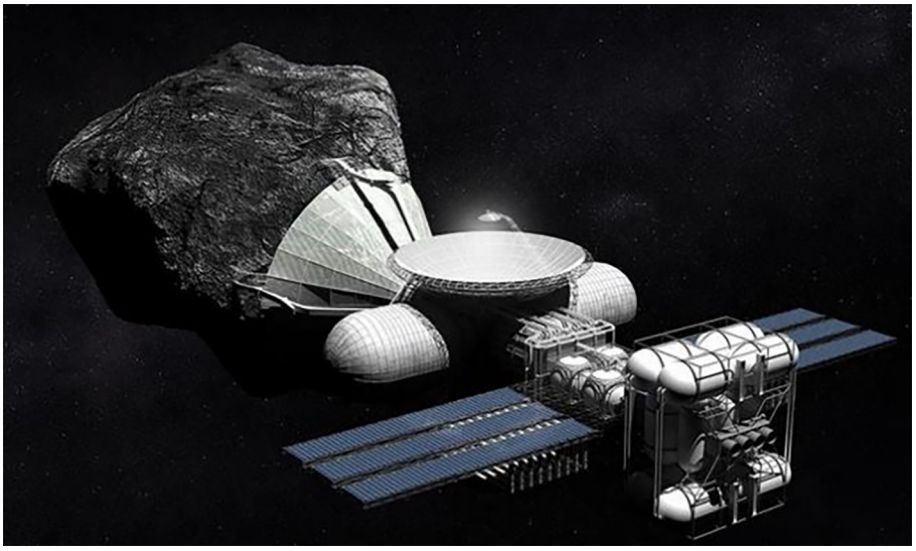


Fig. 15. Illustration of an NEO mining station. This image is taken from a web page [184].

mechanical properties. Methods such as nanoindentation and AFM have become essential. Combined with mineralogical and microstructural analysis, they provide high-precision research tools for studying planetary rocks, greatly enhancing our understanding of these materials and providing a scientific basis for space exploration and related applications.

(2) Microscale mechanical property upscaling methods include homogenization schemes such as the M-T method, the SCS method, the Voigt-Reuss-Hill average method, and the dilute scheme. The RVE homogenization method and numerical simulations like the AGBM are also utilized. Among these, AGBM is more accurate as it incorporates detailed microstructural and mechanical property data, but it currently operates only in 2D. 3D numerical models of meteorites are challenging due to complex microstructures, high computational cost, and nonlinear material behaviors, necessitating advanced damage mechanics model, modeling techniques and further research for comprehensive understanding and practical applications in space missions.

(3) Mineralogical and microstructural analysis methods, along with non-destructive microscale mechanical testing, have certain limitations and influencing factors. Limitations include potential sample damage in some testing methods, the need for specialized equipment, and the challenge of accurately representing complex material heterogeneity. Influencing factors encompass surface roughness, indentation depth, loading rate, and the choice of indenter tips. To reduce these effects, sample preparation is crucial, including polishing to reduce surface roughness and careful selection of testing parameters such as load and depth. It is recommended that check the surface flatness using imaging techniques before micro-RMEs to improve the accuracy of mechanical property assessment.

(4) Current mechanical testing of meteorites is not fully comprehensive, with a greater number of macroscale mechanical tests compared to microscale ones, and some meteorite types lack mechanical data. The limited availability of meteorite samples and the challenges associated with conducting mechanical tests on these precious materials have restricted the extent of testing. To build a reliable dataset, future work should focus on conducting more mechanical tests, improving the data quality, and making a serious testing standard for different types of meteorites, particularly microscale tests, to better understand the intrinsic properties of meteorites and to establish a comprehensive database that can inform space exploration and planetary defense strategies.

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